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REVIEW

A survey of theory-grounded interpretability for deep neural networks

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Abstract

Deep neural networks achieve state-of-the-art performance across computer vision, natural language processing, and scientific discovery, yet their internal decision-making mechanisms remain difficult to interpret. Post-hoc explanation methods such as SHAP, saliency maps, and attention visualization provide empirical insights but offer limited mathematical guarantees about learning dynamics and representation formation. This survey takes a complementary, theory-grounded perspective, examining interpretability through four classical machine learning (ML) frameworks, each grounded in a mature mathematical theory. These are kernel methods (functional analysis), sparse representations (convex optimization), matrix factorization (linear algebra), and manifold learning (differential geometry). We synthesize and critically evaluate 203 papers (2016–2026), organizing them by the theoretical principle each framework contributes rather than by application domain. Neural tangent kernel (NTK) methods provide rigorous convergence and generalization guarantees but depend on idealized infinite-width assumptions; sparse representation methods yield interpretable-by-design architectures but face reconstruction and approximation trade-offs; matrix factorization reveals an implicit bias toward low-rank structure that helps explain generalization; and manifold learning offers geometric tools for analyzing representation structure, class separability, and generative behavior. Beyond reviewing each framework, we establish explicit mathematical bridges between them, arguing that they offer complementary views of a common phenomenon, namely that gradient-based optimization often drives deep networks toward low-dimensional, interpretable structure. We further situate these frameworks among adjacent interpretability paradigms, including mechanistic, concept-based, and attribution-based methods, and delineate the validity conditions and failure modes that determine when classical theory yields reliable rather than misleading explanations. The review concludes with open problems and directions for scalable, theory-grounded interpretability.

Keywords Neural network interpretability · Deep learning theory · Neural tangent kernel · Sparse representations · Matrix factorization · Manifold learning

1 Introduction

Deep learning now achieves state-of-the-art performance across computer vision (He et al. 2016), natural language processing (Vaswani et al. 2017), and scientific discovery (Jumper et al. 2021). Yet this empirical success is accompanied by fundamental opacity. Models function as black boxes whose decision-making processes remain unexplainable. The black-box problem has evolved from a theoretical curiosity into a critical barrier for high-stakes applications that require explanations (Rudin 2019; Lipton 2018). It is particularly concerning because deep neural networks exhibit unexpected behaviors such as vulnerability to adversarial examples (Szegedy et al. 2014), where imperceptible input perturbations can arbitrarily change predictions. These vulnerabilities suggest their underlying mechanisms may differ fundamentally from human reasoning (Zhang et al. 2021b). The goal has therefore shifted from merely achieving high accuracy to enabling transparent knowledge discovery, so that deep learning can serve as a trustwor-

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thy component in systems that require explainable decision support.

This opacity poses the greatest risk in safety-critical applications, where erroneous predictions result in immediate physical or legal harm. In medical diagnosis, for example, deep learning models detect diseases from radiological images with impressive accuracy. However, they cannot extract and validate the diagnostic knowledge needed for clinical decision support, which creates severe trust and liability issues (Shen et al. 2017; Barredo Arrieta et al. 2020; Ennab and Mcheick 2024; Sadeghi et al. 2024). A 2024 NHTSA investigation provides a further example. Federal regulators identified 29 fatal crashes involving Tesla's Autopilot system among 956 analyzed crashes (National Highway Traffic Safety Administration 2024), and limited transparency can hinder both accident investigation and system improvement (Koopman and Wagner 2017). Beyond safety and trust, reliable knowledge extraction serves several other purposes. These include detecting bias in training data, revealing adversarial vulnerabilities, and ensuring that meaningful causal relationships drive predictions (Barredo Arrieta et al. 2020).

The interpretability crisis has attracted significant regulatory attention, leading to binding legal frameworks that mandate explainability. The European Union's Artificial Intelligence (AI) Act, finalized in 2024, establishes the world's first comprehensive AI transparency framework. It carries penalties of up to €35 million or 7% of global annual turnover, whichever is higher (European Parliament and Council of the European Union 2024). Similar regulations have emerged in the United States through NIST's AI risk management framework and sector-specific requirements (National Institute of Standards and Technology 2023; Food 2024; Consumer Financial Protection Bureau 2023; U.S. Equal Employment Opportunity Commission 2023). These regulations respond to critical needs in safety, fairness and non-discrimination, and scientific discovery (Zhang et al. 2021b). Many current post-hoc explanation methods provide limited formal guarantees about global fidelity, robustness, or completeness, so they struggle to satisfy these requirements. This shortfall is most acute for knowledge-based systems, where decision support must be both interpretable and mathematically verifiable. The result is an urgent demand for theoretically grounded approaches.

Following common practice in the ML community (Barredo Arrieta et al. 2020; Lipton 2018), we acknowledge that *interpretability* and *explainability* are closely related concepts that are often used interchangeably to denote the broader goal of understanding how models make predictions. This review focuses on how classical ML theoretical frameworks (kernel methods, sparse representations, matrix factorization, and manifold learning) provide mathematically rigorous foundations for deep learning interpretability. These frameworks differ fundamentally from widely used post-hoc methods (SHAP, LIME, attention visualization), which explain individual predictions of a trained model without accounting for the learning dynamics that produce its representations. Classical frameworks enable both the analysis of trained networks through rigorous mathematical tools and the construction of interpretable-by-design architectures in which interpretability is built into the network structure from inception. This review makes explicit distinctions between these approaches where they matter for understanding the methods and their validity conditions.

Beyond these four frameworks, the interpretability landscape includes paradigms such as mechanistic interpretability and circuit analysis (Räuker et al. 2023), concept-based explanations (Kim et al. 2018), causal interpretability (Geiger et al. 2022), and axiomatic attribution (Sundararajan et al. 2017). These directions are often presented as alternatives to classical theory or grouped loosely with post-hoc methods, but this characterization is incomplete. Several of the most rigorous instances of these paradigms, such as sparse autoencoders in mechanistic interpretability and some concept-based explanation methods, rest directly on the classical foundations surveyed here. Others, such as causal interpretability and axiomatic attribution, are mathematically rigorous but rest on different foundations, namely causal inference and axiomatic or game-theoretic principles, and address a different question from how networks learn and form representations. Drawing these distinctions clearly is part of what this review contributes. We therefore make our scope boundary explicit in Section 1.3.1. There we separate the directions that fall naturally within our four frameworks from those that constitute distinct research domains.

1.1 Research gap

In response to the pressures described above, the ML community has developed numerous post-hoc explanation methods, including SHAP (Lundberg and Lee 2017), LIME (Ribeiro et al. 2016), saliency maps (Simonyan et al. 2014), and attention visualizations (Clark et al. 2019), reviewed in recent general and domain-specific surveys (Barredo Arrieta et al. 2020; Zhang et al. 2021b; Li et al. 2022; Ennab and Mcheick 2024; Xu and Yang 2025; Fridkin and Bendersky 2026). However, many post-hoc explanations have been criticized for their vulnerability to adversarial manipulation (Slack et al. 2020), their failure to pass basic reliability tests (Adebayo et al. 2018), and the absence of universally accepted evaluation standards. They often provide feature-importance, sensitivity, or local-surrogate summaries rather than mechanistic accounts of how deep networks learn and form representations. Although some rest on rigorous foundations under specific assumptions, they generally offer no formal guarantees of global correctness, completeness, robustness,

or fidelity to the full internal computation of the underlying model.

In contrast, classical ML frameworks offer mature mathematical tools and, in certain settings, provable guarantees, enabling more principled interpretability analysis. Table 1 compares our review with recent interpretability surveys. Surveys exist for individual interpretability paradigms and for application-specific uses of classical techniques, includ-

Table 1 Comparison of recent deep learning interpretability surveys

Survey	Focus	Key Coverage	Classical ML?
Early Foundational Surveys (2018-2020)			
Gilpin et al. (2018)	Explanations overview	Interpretability definitions, desiderata, complete vs incomplete explanations	Not covered
Murdoch et al. (2019)	Conceptual framework	Interpretability definitions, applications, PDR evaluation	Not covered
Carvalho et al. (2019)	Methods and metrics	Quantitative evaluation metrics, model-specific methods	Not covered
Barredo Arrieta et al. (2020)	Comprehensive explainable artificial intelligence (XAI)	LIME, SHAP, attention mechanisms, visualization, responsible AI	Not covered
Linardatos et al. (2021)	Interpretability methods	Model-agnostic methods, model-specific methods, evaluation approaches	Not covered
Domain-Specific Surveys (2021-2022)			
Zhang et al. (2021b)	Neural network interpretability	Method categorization, evaluation metrics, reliability dimensions	Not covered
Li et al. (2022)	General interpretability	Interpretation taxonomy, evaluation metrics	Not covered
Recent Comprehensive Surveys (2023-2026)			
Saeed and Omlin (2023)	Meta-survey of XAI	Systematic review of challenges, future opportunities, research gaps	Not covered
Räuker et al. (2023)	Inner structures of deep neural networks (DNNs)	Mechanistic interpretability, internal representations, circuit analysis	Brief mention only
Hosain et al. (2024)	XAI in deep learning	Applications across domains, XAI approaches, evaluation metrics	Not covered
Hassija et al. (2024)	Black-box models	LIME/SHAP, trust, regulatory compliance, bias detection	Not covered
Xu and Yang (2025)	Classic and modern methods	Post-hoc explanations, visualization, attention, causality	Brief mention only
Shu et al. (2025)	Sparse autoencoders for LLM interpretability	SAE architectures, feature explanation, evaluation, mechanistic interpretability of LLMs	Brief mention only
Fridkin and Bendersky (2026)	Comprehensive Interpretable ML foundations and methods	Intrinsic models, post-hoc model-agnostic methods, deep learning attribution, mechanistic interpretability	Brief mention only
Hu et al. (2026)	Interpretable clustering	Pre-/in-/post-clustering interpretability, decision trees, rules, prototypes	Not covered
Application-focused surveys (for contrast):			
Zhang et al. (2024a)	Hyperspectral imaging (HSI) denoising performance	Sparse/low-rank methods as algorithmic tools	*Used as algorithms, not theory*
This Review	**Theoretical foundations**	**Systematic analysis of classical ML theory**	**Unified theoretical treatment across four frameworks**

ing reviews that apply sparse and low-rank methods as algorithmic tools (Zhang et al. 2024a), as well as recent surveys focused on individual paradigms such as sparse autoencoders for language-model interpretability (Shu et al. 2025). To the best of our knowledge, however, no existing survey systematically examines kernel methods, sparse representations, matrix factorization, and manifold learning as a unified set of classical theoretical foundations for deep learning interpretability. Nor does any existing survey connect these frameworks to one another through their shared role in explaining learning dynamics and representation formation. Figure 1 presents our comprehensive taxonomy, which organizes these frameworks and the 203 papers reviewed in this work.

Recent work has demonstrated the practical relevance of these classical frameworks at modern deep-learning scale. Sparse autoencoders have been used to extract millions of interpretable features from frontier and open language models (Templeton et al. 2024; Lieberum et al. 2024) (Section 3); NTK-based methods provide tools for analyzing linearized training dynamics, convergence, and kernel-based surrogate explanations (Jacot et al. 2018; Engel et al. 2024) (Section 2); and matrix factorization methods decompose transformer activations into interpretable components (Shafran et al. 2026) (Section 4). Sections 2–5 systematically review these and related manifold-learning approaches, examining their theoretical foundations, practical applications, validity conditions, and limitations.

Despite this progress, the theoretical foundations underlying these methods remain fragmented across research communities. Each framework addresses different aspects of interpretability under distinct assumptions. Kernel theory often relies on idealized infinite-width or linearized regimes, sparse methods face reconstruction and feature faithfulness

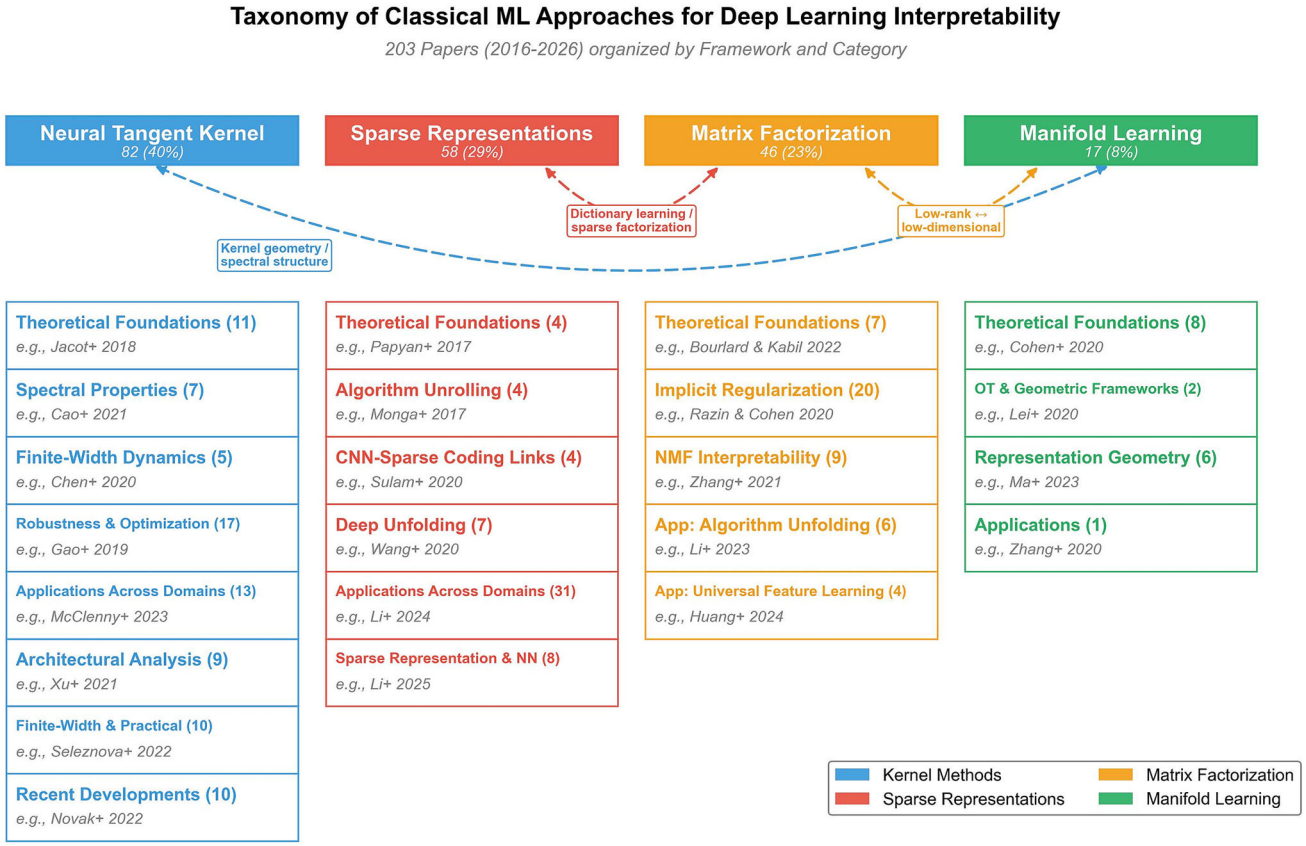


Fig. 1 Comprehensive taxonomy of classical ML approaches for deep learning interpretability. The 203 reviewed papers (2016–2026) are organized hierarchically into four classical ML frameworks: kernel methods (82 papers, 40%), sparse representations (58 papers, 29%), matrix factorization (46 papers, 23%), and manifold learning (17 papers, 8%). Each framework is subdivided into different categories based on theoretical focus and application domains, with representative exemplar papers shown. The distribution reveals that kernel methods dominate due to extensive NTK theoretical developments, while manifold learning remains relatively underexplored despite its theoretical potential. Dashed arrows indicate the principal cross-framework links: sparse representations connect to matrix factorization through dictionary learning and sparse factorization; matrix factorization connects to manifold learning through the shared principle of low-dimensional structure; and NTK methods connect to manifold learning through kernel geometry and spectral structure. These connections are expanded into a unified mathematical synthesis in Fig. 8 and analyzed in Section 6.1.3

trade-offs, matrix factorization depends on low-rank or parts-based structure, and manifold methods rely on geometric assumptions about representation spaces. This review fills the resulting gap by providing a unified treatment of these four classical frameworks, systematically analyzing what works, under what conditions, and what barriers prevent complete theoretical understanding of finite-width networks, the regime where practical deep learning operates.

1.2 Motivation and background

Despite deep learning's empirical success (Bartlett et al. 2021; Roberts et al. 2022), a complete mathematical theory is still lacking to guide model improvement (Xu and Yang 2025), which contributes to trial-and-error tuning and trust deficits, since statistical performance metrics alone provide insufficient evidence of trustworthiness (Scorzato 2024). This theoretical gap motivates our focus on classical ML frameworks, which offer mature mathematical tools and, in certain settings, provable guarantees across four families. First, kernel methods connect learning to functional analysis in reproducing kernel Hilbert spaces and provide theoretical guarantees about convergence and generalization (Schölkopf and Smola 2002; Vapnik 1999). Second, sparse representations draw on sparse approximation, convex relaxation, and dictionary learning to decompose signals into interpretable basis elements, where sparsity provides explicit and quantifiable interpretability (Donoho 2006; Elad and Aharon 2006). Third, matrix factorization methods apply linear algebra to reveal latent structure through low-rank and parts-based decompositions (Lee and Seung 1999; Golub and Van Loan 2013). Fourth, manifold learning characterizes how high-dimensional data concentrate on low-dimensional structures (Tenenbaum et al. 2000; Roweis and Saul 2000).

Despite their theoretical strengths, these frameworks have often been applied to deep learning interpretability in isolation across different research communities. Kernel theorists investigate NTKs (Jacot et al. 2018) largely independently of researchers applying sparse coding to neural networks (Cunningham et al. 2024), while work on matrix factorization (Shafran et al. 2026) and manifold-based analyses of representation geometry (Cohen et al. 2020) often proceed with limited connection. Many classical ML methods offer clearer mathematical structure, but in many modern high-dimensional tasks they have not matched deep networks' empirical performance on their own. Conversely, deep networks perform well but resist interpretation. Realizing the interpretability benefit therefore requires understanding how each classical framework connects to deep networks and under what conditions it remains valid. Table 8 synthesizes their complementary strengths through comparative analysis.

1.3 Scope and objectives

This review addresses five fundamental questions about applying classical ML theory to deep learning interpretability, as follows:

1. How can kernel methods, sparse representations, matrix factorization, and manifold learning provide rigorous frameworks for understanding neural networks?
2. What are the theoretical foundations, assumptions, and validity conditions for each classical framework?
3. Under what circumstances does each approach provide genuine interpretability insights rather than misleading or incomplete explanations?
4. How do different classical approaches compare in terms of theoretical rigor, practical applicability, and validity conditions?
5. What fundamental limitations prevent current theoretical approaches from fully explaining finite-width deep networks operating in the rich feature learning regime?

This review excludes (1) post-hoc explanation methods that explain trained-model outputs rather than how networks learn and form representations (SHAP, LIME, CAM variants), (2) alternative interpretable models that do not provide mechanistic understanding of neural network behavior, and (3) purely empirical studies without generalizable theoretical insights. The four frameworks examined in this survey were selected because each provides a mathematically distinct, self-contained lens on neural network behavior, drawing respectively on functional analysis, convex optimization, linear algebra, and differential geometry. Together they span the principal axes along which deep networks have been rigorously analyzed, namely training dynamics, representational structure, implicit bias, and geometric organization, while remaining individually tractable. Section 1.3.1 discusses how the excluded methods and other adjacent paradigms relate to these four frameworks, distinguishing directions that intersect them from those that rest on different foundations.

1.3.1 Relation to adjacent interpretability paradigms

Several active interpretability directions lie beyond our four frameworks, including mechanistic interpretability and circuit analysis, concept-based explanations (Kim et al. 2018), causal interpretability, axiomatic and game-theoretic attribution, polynomial or Taylor-expansion explanations (Montavon et al. 2017), information-theoretic analyses (Tishby and Zaslavsky 2015), spectral-graph approaches (Defferrard et al. 2016), and symbolic or rule-extraction methods (Padalkar and Gupta 2025). Many of these rest on genuine mathematical foundations rather than heuristic reasoning,

and we do not dismiss them. Several intersect our frameworks directly, and we cover that intersection where it arises. For instance, the sparse autoencoders now widely used in mechanistic interpretability can be understood as instances of sparse dictionary learning (Section 3) (Cunningham et al. 2024; Li et al. 2025); some automatic and invertible concept-based explanation methods obtain concept directions through matrix factorization, including non-negative matrix factorization of intermediate representations (Section 4) (Zhang et al. 2021a); and symbolic rules have been extracted from sparse concept representations of vision transformers (Padalkar and Gupta 2025).

Other directions are rigorous but rest on foundations outside the four classical theories we organize around, and address a different question. Causal interpretability (Geiger et al. 2022) is grounded in structural causal models and interventions, while axiomatic attribution such as Integrated Gradients (Sundararajan et al. 2017) rests on an axiomatic path-integral framework and SHAP (Lundberg and Lee 2017) on cooperative game theory, and polynomial or Taylor-expansion explanations (Montavon et al. 2017) provide principled local decompositions of a trained model's predictions. The attribution and Taylor methods primarily explain the outputs of a fixed trained model, while causal interpretability asks how representations relate to an underlying causal structure; none of these provides the classical learning-theoretic account of how networks learn and form representations that this review targets. These paradigms are therefore complementary to our scope rather than missing from it.

1.4 Methodology

This review adopts a systematic search methodology informed by PRISMA 2020 reporting principles to ensure transparency and reproducibility in literature selection (Page et al. 2021). The complete selection process is summarized in the PRISMA flow diagram of Fig. 2. A comprehensive literature review was conducted using three databases. These are Web of Science (WoS) Core Collection, Scopus, and IEEE Xplore, chosen because together they capture the ML and signal processing communities in which the four frameworks are developed, and because they index peer-reviewed venues and provide reliable citation counts for the threshold-based screening described below (Mongeon and Paul-Hus 2016). Publications from 2016–2026 were searched using the following query:

(*"neural tangent kernel"* OR *"NTK"* OR *"sparse coding"* OR *"sparse representation"* OR *"dictionary learning"* OR *"matrix factorization"* OR *"NMF"* OR *"manifold learning"* OR *"implicit regularization"* OR *"kernel*

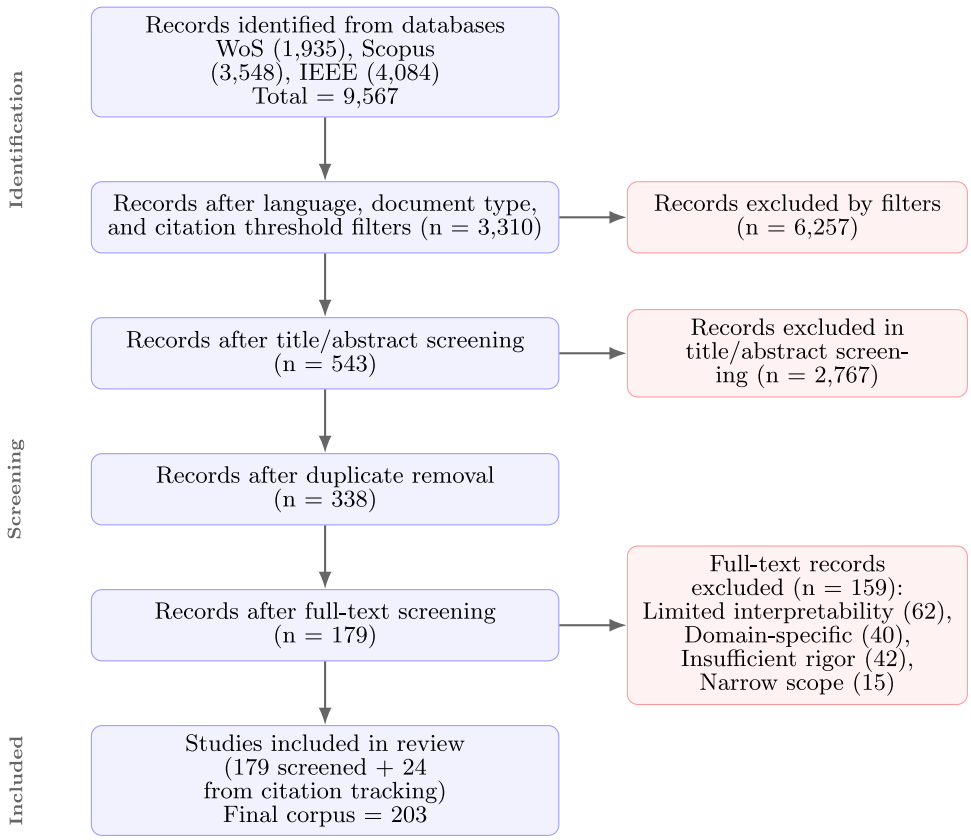


Fig. 2 PRISMA-style flow diagram of the study selection process. Counts correspond to Table 2

regime" OR *"deep linear network"* OR *"infinite-width"*) AND (*"deep learning"* OR *"neural network"*) AND (*interpret** OR *explain** OR *understanding* OR *transparent** OR *analysis*)

The final database search was conducted on 28 March 2026. WoS was searched within Topic fields yielding 1,935 results; Scopus was searched within Article Title/Abstract/Keywords across Computer Science, Engineering, and Mathematics subject areas yielding 3,548 results; IEEE Xplore was searched across All Metadata yielding 4,084 results. After applying language, document type, and citation threshold filters, 456 (WoS), 1,233 (Scopus), and 1,621 (IEEE) papers remained (3,310 combined). The same differential citation thresholds were applied across all three databases, with papers from 2016–2022 required to have $\geq$11 citations and papers from 2023–2026 $\geq$3 citations.

Title and abstract screening was performed within each database separately, reducing the set to 117 (WoS), 301 (Scopus), and 125 (IEEE) papers (543 combined). Because screening was conducted per database, duplicate records appearing in more than one database were removed at the merge stage, and after removing 205 such duplicates identified through both title and abstract matching, 338 unique papers remained. Full-text screening against inclusion and exclusion criteria retained 179 papers. An additional 24 papers were identified through backward and forward citation tracking of retained papers. The final corpus comprises **203 papers** for detailed review (Table 2).

Because the review targets theory-grounded interpretability rather than applications, screening retained only papers that use a classical framework to explain how or why networks learn and behave, and excluded those that use it merely as a performance tool. Inclusion criteria required papers applying classical ML theoretical frameworks to deep learning interpretability with mathematical rigor and mechanistic understanding. Exclusion criteria eliminated purely empirical applications without theoretical insights, post-hoc explanation methods outside the four classical frameworks (Section 1.3.1), and work focused solely on performance improvement rather than interpretability. Papers were excluded under limited interpretability when they applied a classical framework solely to improve performance, for example optimization speed or predictive accuracy, without using it to explain or mechanistically understand neural network behavior.

Figure 1 presents the comprehensive taxonomy organizing these 203 papers into four classical ML frameworks and 23 subcategories based on theoretical foundations and application domains. The distribution reveals kernel methods as the dominant framework (40%), driven primarily by extensive NTK theoretical developments since 2018, followed by sparse representations (29%), matrix factorization (23%), and manifold learning (8%). This organizational structure guides the remainder of the paper, with Sections 2–5 systematically reviewing each framework.

Each framework is examined through three complementary lenses. These are (i) theoretical rigor (mathematical

Table 2 Systematic screening process from initial search to final selection informed by PRISMA 2020 reporting principles. Three databases were searched with differential citation thresholds to balance established and emerging work. Full-text screening applied inclusion/exclusion criteria focused on interpretability contribution, theoretical rigor, and generalizability

Stage	WoS	Scopus	IEEE	Combined	Action
Initial Search	1,935	3,548	4,084	9,567	Database queries
After Filters	456	1,233	1,621	3,310	Language, document type, citation threshold
Title/Abstract Screening	117	301	125	543	Remove irrelevant papers
After Deduplication	–	–	–	338	205 duplicates removed
Full-Text Screening	–	–	–	179	Apply inclusion/exclusion criteria
Exclusion reasons (159 papers removed):					
Limited interpretability	–	–	–	62	Optimization focus only
Domain-specific	–	–	–	40	No generalizable theory
Insufficient rigor	–	–	–	42	Lacks mathematical foundations
Narrow scope	–	–	–	15	Single phenomenon only
Citation Tracking	–	–	–	+24	Backward/forward search
Final Corpus	–	–	–	**203**	**For detailed review**
Framework Distribution					
Neural Tangent Kernel	–	–	–	82 (40%)	
Sparse Representations	–	–	–	58 (29%)	
Matrix Factorization	–	–	–	46 (23%)	
Manifold Learning	–	–	–	17 (8%)	

Citation thresholds: 2016–2022 papers required $\geq$11 citations (Scopus/WoS/IEEE); 2023–2026 papers required $\geq$3 (Scopus/WoS/IEEE)

foundations, formal guarantees, validity conditions), (ii) practical applicability (computational feasibility, scalability to modern architectures), and (iii) interpretability insights (mechanistic understanding versus statistical correlation, faithfulness to model behavior). This multi-faceted analysis enables comparative assessment of complementary strengths and fundamental limitations across frameworks.

Foundational works cited for theoretical context were exempt from citation thresholds. A work was classified as foundational if it (i) introduced a theoretical framework or mathematical formalism central to one of the four reviewed frameworks (e.g., the original NTK formulation, the sparse coding model, matrix factorization, the manifold hypothesis), (ii) was published before the 2016 review window, and (iii) is required for expository completeness rather than as evidence of a recent trend. Examples include (Olshausen and Field 1996; Lee and Seung 1999; Tenenbaum et al. 2000), and Daubechies et al. (2004).

In addition, a small number of further references (Li et al. 2026; Martins and Astudillo 2016; Correia et al. 2019; Treviso et al. 2022; Simon and Zou 2025) were added during revision to address reviewer comments, together with references cited only for scope positioning in Section 1.3.1 and Table 1. Like the foundational works, these references were not subject to the screening criteria and are not counted in the corpus of 203 papers.

1.5 Limitations of the review methodology

While this comprehensive approach ensures transparency and reproducibility, several methodological limitations require acknowledgment. First, database coverage is restricted to WoS, Scopus, and IEEE Xplore, which may miss a small number of peer-reviewed papers indexed primarily in the ACM Digital Library or DBLP, although much computer-science work in these indexes is also covered by Scopus. This choice prioritizes established, citation-indexed venues, and non-peer-reviewed preprints were excluded by design, consistent with a focus on peer-reviewed publications.

Second, the citation thresholds ($\geq$11 citations for papers from 2016–2022 and $\geq$3 for papers from 2023–2026, applied uniformly across all three databases) introduce a potential bias toward established work. These thresholds were used as a pragmatic relevance filter to keep the corpus tractable, rather than as a measure of intrinsic paper quality. The two-tier design reflects citation accrual time. Papers from 2016–2022 have had several years to accumulate citations, so a higher threshold screens for demonstrated community uptake, whereas papers from 2023–2026 are held to a lower threshold so that recent but under-cited work is not systematically excluded.

Although the lower threshold for recent papers reduces this effect, citation-based filtering inevitably disadvantages the newest work, since recent papers have had less time to accumulate citations. Backward and forward citation tracking partly offsets this by recovering relevant recent papers that the threshold-filtered search missed. No single threshold perfectly balances quality (favoring highly cited work) with recency (including recent contributions).

Third, the exclusion criteria based on theoretical rigor and interpretability contribution involve subjective judgment. Although consistent screening protocols were applied, different researchers might classify borderline papers differently. This review explicitly focuses on the four frameworks of NTK, sparse representations, matrix factorization, and manifold learning, a choice that excludes certain valid interpretability approaches.

1.6 Novel contributions

This review makes the following contributions:

1. A comprehensive review connecting classical ML theory (kernel methods, sparse representations, matrix factorization, manifold learning) to deep learning interpretability, synthesizing 203 papers (2016–2026).
2. A unified framework that organizes methods by epistemological foundation rather than application domain, revealing fundamental relationships between approaches that transcend specific use cases. We further establish explicit mathematical bridges between the four frameworks, arguing that they offer complementary views of a common underlying tendency in how gradient-based optimization shapes deep networks.
3. A critical evaluation of theoretical rigor versus practical applicability, exposing the trade-off between methods with strong mathematical guarantees (kernel methods) and those with practical utility (sparse approaches).
4. A validity analysis that establishes when and why classical frameworks provide genuine interpretability insights rather than misleading or incomplete explanations for finite-width networks.
5. The identification of critical open problems where current theoretical approaches remain insufficient, providing concrete directions for future research.
6. A methodological critique that articulates weaknesses in current approaches, including unrealistic assumptions (infinite-width limits), lack of finite-width theory, and gaps between theory and practice.

1.7 Organization of the paper

The remainder of this paper is organized as follows. Section 2 examines kernel methods, particularly the NTK, for understanding neural network training dynamics. Section 3 explores sparse representation and algorithm unrolling

approaches and demonstrates how classical optimization algorithms can be transformed into interpretable deep architectures. Section 4 investigates matrix factorization methods and examines implicit regularization in deep networks and NMF-based interpretability approaches. Section 5 examines manifold learning perspectives on neural representations and geometric transformations in deep networks. Section 6 concludes with a synthesis of key findings, a critical analysis of fundamental limitations, and recommendations for future theoretical developments.

2 Neural tangent kernel methods

The NTK framework represents one of the most significant theoretical advances in understanding deep learning through classical kernel methods. This framework establishes a rigorous connection between neural network training dynamics and kernel gradient descent. It enables mathematical analysis of phenomena previously understood only empirically, including convergence guarantees, generalization properties, and the implicit regularization of gradient descent.

2.1 Theoretical foundations

The NTK framework, introduced by Jacot et al. (2018), provides a rigorous theoretical lens for understanding deep learning through kernel methods. This framework is particularly significant because it enables global convergence guarantees and interpretable learning dynamics for neural networks, which is something previously unavailable for general deep learning systems. Deep neural networks are trained using gradient descent or its variants such as stochastic gradient descent (SGD) and adaptive moment estimation (Adam) (Kingma and Ba 2015). Understanding the training dynamics therefore requires analyzing how gradient descent updates affect the network's predictions. Jacot et al. demonstrated that gradient descent optimization in parameter space is mathematically equivalent to kernel-based gradient flow in function space, where the kernel in question is the NTK. More formally, consider a fully-connected neural network of depth L with width m at each layer. The NTK at random initialization $\theta_0 \sim \mathcal{N}(0, I)$ is defined as

$$\Theta^{(L)}(x, x') = \sum_{p=1}^{P} \frac{\partial f(x;\theta_0)}{\partial \theta_p} \frac{\partial f(x';\theta_0)}{\partial \theta_p} \tag{1}$$

as illustrated in the middle box of Fig. 3, where $P = \sum_{\ell=0}^{L-1}(n_\ell+1)n_{\ell+1}$ is the total number of parameters (weights and biases) in the network, $\theta_0 = \{W^{(\ell)}, b^{(\ell)}\}_{\ell=0}^{L-1}$ is the full parameter vector at initialization, collecting all weight matrices $W^{(\ell)} \in \mathbb{R}^{n_\ell \times n_{\ell+1}}$ and bias vectors $b^{(\ell)} \in \mathbb{R}^{n_{\ell+1}}$, with each scalar parameter drawn independently from $\mathcal{N}(0, 1)$, θ_p denotes the p-th individual scalar parameter obtained by enumerating all elements of θ_0, and x, x' are any two input examples to the network with $f(x;\theta_0) \in \mathbb{R}^{n_L}$ denoting the network output at input x under parameters θ_0. For instance, in image classification, x might be a photo of a cat and x' a photo of a dog, with the kernel $\Theta^{(L)}(x, x')$ quantifying how the network's parameter gradients couple its responses to these two images. This transformation from parameter space to function space optimization provides several key advantages as shown in Table 3, most notably converting non-convex optimization into convex optimization with global convergence guarantees.

The key theoretical result establishing this framework considers the infinite-width limit where hidden layer widths $n_1, \ldots, n_{L-1}$ grow to infinity, keeping input dimension n_0 and output dimension n_L fixed. In this limit, remarkable simplifications occur that make the training dynamics fully analyzable.

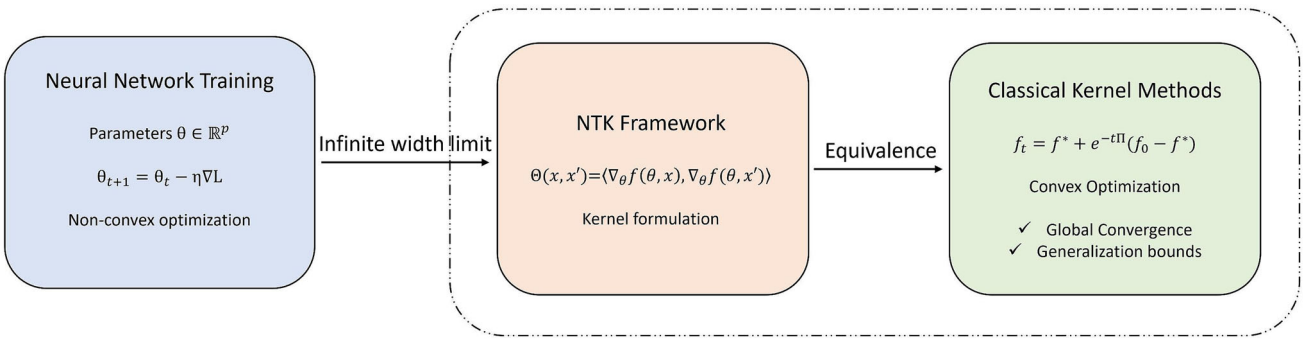


Fig. 3 The NTK framework establishes mathematical equivalence between neural network training and kernel regression through a two-step transformation. First, in the infinite-width limit ($m \to \infty$), neural network training dynamics are captured by the NTK framework. Second, the NTK framework is mathematically equivalent to classical kernel regression. This transforms non-convex optimization in parameter space into convex optimization in function space with theoretical guarantees for convergence and generalization. These guarantees hold strictly under the infinite-width lazy regime, least-squares loss, gradient flow dynamics, and positive definiteness of the limiting NTK

Table 3 Comparison of parameter space versus function space (NTK) optimization properties

Property	Parameter Space View	Function Space View (NTK)
Optimization Type	Non-convex	Convex†
Loss Landscape	Many local minima	Single global minimum†
Saddle Points	Present	Absent
Space	$\theta \in \mathbb{R}^P$	$f \in \mathcal{F}$
Convergence	No guarantees	Guaranteed global convergence†
Analysis	Intractable	Tractable (kernel methods)
Interpretability	Opaque dynamics	Kernel PCA, spectral analysis

†Function-space convexity, single global minimum, and guaranteed convergence hold conditionally on positive definiteness of the limiting NTK, a convex and bounded-below loss, and the infinite-width lazy regime. These guarantees break down for finite-width networks in the rich feature-learning regime

Theorem 1 (Jacot et al. 2018) *For a fully-connected network with Lipschitz nonlinearity σ, as the layer widths $n_1, \ldots, n_{L-1} \to \infty$, the NTK $\Theta_m^{(L)}(x, x')$ converges in probability to a deterministic kernel $\Theta_\infty^{(L)}(x, x')$ that depends only on the depth L, activation function, and initialization variance (Theorem 1 of Jacot et al. 2018). Moreover, during training with gradient descent on any fixed time horizon $t \in [0, T]$, the NTK remains asymptotically constant in the infinite-width limit: $\Theta^{(L)}(t) \to \Theta_\infty^{(L)} \otimes I_{n_L}$ uniformly in t (Theorem 2 of Jacot et al. 2018).*

The mechanism underlying this constancy was characterized by Liu et al. (2020), who proved that the transition to linearity of the model, and equivalently the constancy of the NTK, results from the scaling properties of the Hessian matrix norm as a function of network width. They showed that for architectures with a linear output layer, the Hessian spectral norm vanishes as $\tilde{O}(1/\sqrt{m})$ while the gradient remains $\Theta(1)$. This forces the model to behave linearly in a ball of fixed radius. They also explicitly demonstrated that this does not hold universally for all wide networks. Subsequent work by Woodworth et al. (2020) revealed a complementary route into the lazy regime. Under standard parameterization, a fundamental dichotomy in training behavior is controlled by the initialization scale α. This complements the Jacot et al. analysis, which relies on NTK parameterization where infinite width alone induces lazy behavior regardless of α. The lazy (kernel) regime can be reached either via NTK parameterization with $m \to \infty$ or via standard parameterization with $\alpha \to \infty$. In this regime, network parameters change minimally during training. The NTK remains frozen at initialization ($\Theta(t) \approx \Theta(0)$), causing the network to behave as kernel regression with fixed features confined to the reproducing kernel Hilbert space (RKHS) of the initial kernel. This regime enables exact theoretical predictions and convergence guarantees but limits representational power and generalization performance.

Conversely, in the rich (feature learning) regime achieved through small initialization scale ($\alpha \to 0$), all layers adapt significantly during training. The NTK evolves ($\Theta(t) \neq \Theta(0)$), enabling the network to learn hierarchical representations such as sparse or low-rank structures. These representations go beyond the initial RKHS. Woodworth et al. proved that for diagonal linear networks of any depth $L \geq 2$, this rich limit induces an ℓ_1 norm minimization bias. This contrasts with the ℓ_2 norm bias of the lazy regime. While this rich regime enables the superior performance underlying modern deep learning's success, theoretical guarantees break down and rigorous analysis becomes intractable. Under standard parameterization, the initialization scale acts as the primary control parameter for this transition. Large α forces lazy dynamics while small α enables the feature learning that drives practical effectiveness. Under NTK parameterization, infinite width alone induces lazy behavior irrespective of α. Geiger et al. (2020) provided a complementary analysis showing that generalization error does not increase monotonically with parameter count as classical bias-variance arguments would suggest. Instead, generalization error follows a double-descent-like scaling pattern consistent with NTK interpolation behavior. The fundamental tension between theoretical tractability and representational power is detailed in Table 4.

The constant NTK property established by Theorem 1 enables analysis in the space of functions computed by the network rather than in the space of network weights. In this regime, network outputs behave like kernel regression with the NTK, connecting deep learning to Gaussian processes and classical kernel theory. This transformation reframes the originally non-convex optimization in parameter space as kernel gradient descent in function space, where the squared-loss objective is convex with respect to the learned function f. The parameters themselves continue to evolve non-convexly; convexity is obtained only in the function-space view and only for losses that are convex in f (e.g., least-squares regression). Specifically, for a least-squares regression loss defined as the mean squared error $\mathcal{C}(f) = \frac{1}{2n}\sum_{i=1}^{n}(f(x_i) - y_i)^2$, and under gradient flow

Table 4 Comparison of lazy (kernel) and rich (feature learning) training regimes. Jacot et al. (2018) derived the lazy regime under NTK parameterization via the infinite-width limit. Woodworth et al. (2020) characterized the transition under standard parameterization where initialization scale α controls the regime at any width

Training Regime Transition		
Aspect	Lazy Regime (Kernel)	Rich Regime (Feature Learning)
Conditions	*NTK param.*: Infinite width ($m \to \infty$) suffices regardless of α (Jacot et al. 2018). *Standard param.*: Large α ($\alpha \to \infty$) (Woodworth et al. 2020)	*Standard param.*: Small α ($\alpha \to 0$); harder to reach at large width
NTK Behavior	$\Theta(t) \approx \Theta(0)$	$\Theta(t) \neq \Theta(0)$
	Remains constant	Evolves during training
Parameter Change	Minimal movement from initialization	Significant adaptation
Feature Representation	Fixed at initialization	Learned during training
Learning Mechanism	Kernel regression with frozen features	Hierarchical representation learning
Implicit Bias	Min. RKHS norm (reduces to min. ℓ_2 for depth-2 diagonal linear networks under Woodworth et al.'s parameterization (Woodworth et al. 2020); requires gradient flow, squared loss, $\alpha \to \infty$)	Min. ℓ_1 norm in the limit $\alpha \to 0$, proven for diagonal linear networks of any depth $L \geq 2$ (Woodworth et al. 2020); requires *exponentially small* α for $L = 2$, polynomially small for $L \geq 3$; **not established for general nonlinear networks**
Theory	✓ Exact predictions	× Theory breaks down
	✓ Convergence proofs	Limited guarantees
	✓ Tractable analysis (Jacot 2018)	
Performance	Limited by RKHS	Superior generalization
	Weaker generalization	Learns complex features
Practical Relevance	Rare in practice	⋆ **Modern deep learning operates here**
	(very large α or infinite width)	(small/moderate α, finite width)
The Fundamental Tension:	*Theory works ↔ Performance suffers*	*Performance excellent ↔ Theory limited*

dynamics in the infinite-width lazy regime ($\Theta(t) \approx \Theta(0)$), the network's prediction evolves according to

$$\frac{df_t(x)}{dt} = -\frac{1}{n}\sum_{i=1}^{n} \Theta^{(L)}(x, x_i)(f_t(x_i) - y_i) \tag{2}$$

where n is the number of training samples, $(x_i, y_i)_{i=1}^n$ denotes the training dataset with input-output pairs, and $f_t(x_i)$ is the network's prediction at time t for input x_i. For networks with scalar output ($n_L = 1$), $\Theta^{(L)}(x, x_i) \in \mathbb{R}$ and the products are scalar; for vector-valued outputs ($n_L > 1$), $\Theta^{(L)}(x, x_i) \in \mathbb{R}^{n_L \times n_L}$ is the matrix-valued NTK of Theorem 1, and each term $\Theta^{(L)}(x, x_i)(f_t(x_i) - y_i)$ is a matrix-vector product in $\mathbb{R}^{n_L}$. The term $f_t(x_i) - y_i$ is the least-squares residual (following Sections 3 and 5 of Jacot et al. 2018). The negative sign reflects gradient descent on the least-squares loss, driving predictions toward targets y_i. This equation holds strictly under gradient flow on least-squares loss. Extensions to discrete gradient descent or other losses require additional assumptions. Equation 2 is precisely kernel gradient descent with respect to the NTK $\Theta^{(L)}$ defined in (1).

Under least-squares regression, the network's output evolves according to the linear differential (2). The solution can be written as $f_t = f^* + e^{-t\Pi}(f_0 - f^*)$ as shown in the right box of Fig. 3, where f^* denotes the optimal solution (zero training loss), f_0 is the initial network output, and Π is a linear operator determined by the NTK. The learning dynamics converge most rapidly along directions corresponding to the NTK's largest principal components, with each component i converging at rate $e^{-\lambda_i t}$ where λ_i is the corresponding eigenvalue. This mathematical characterization justifies early stopping as a regularization method. Halting training early prioritizes learning the most significant patterns and prevents overfitting to less important directions with smaller eigenvalues. Lee et al. (2020) demonstrated that these eigendirections relate to generalized leverage scores of the training data, proving that leverage score sampling can be generalized to the NTK framework to accelerate training by identifying and prioritizing the most informative training examples. The framework guarantees convergence to a global optimum when the limiting NTK satisfies positive definiteness for the data distribution, directly linking architectural choices to optimization success (Jacot et al. 2018). These guarantees require (i) positive definiteness of the limiting NTK on the training data, (ii) a convex and bounded-below loss (e.g., least-squares regression), and (iii) gradient flow dynamics in the infinite-width limit where $\Theta(t) \approx \Theta_\infty^{(L)}$ remains constant. For finite-width networks operating in

the rich feature-learning regime, these guarantees break down entirely and no analogous theoretical results currently exist. These foundational results were established for fully-connected feedforward networks. Extensions to finite-width networks and other architectures such as CNNs and transformers require more careful analysis, as these extensions may exhibit departures from strict NTK behavior.

The theoretical foundations have been significantly extended by understanding the conditions under which networks can be trained successfully. Xiao et al. (2020) demonstrated that trainability and generalization can be disentangled through NTK spectral analysis. They also showed that CNNs with pooling exhibit fundamentally different NTK spectra than fully-connected networks. Bartlett et al. (2021) synthesized the statistical viewpoint, showing that while NTK theory characterizes interpolation behavior in the overparameterized regime, it misses the rich implicit regularization found in the feature-learning regime. This finding precisely delineates the theory's operating domain. Fang et al. (2021) provided a comprehensive synthesis clarifying when and why neural networks follow NTK theory. Ergen and Pilanci (2021) developed a convex analytic approach for finite-width two-layer ReLU networks, proving global optimization is achievable through convex programming. Yang and Littwin (2021) established that ResNets and transformers exhibit NTK behavior universally in the infinite-width limit throughout training.

2.2 Extensions and dynamics beyond basic NTK theory

The foundational NTK framework has been extended in several important directions that address its limitations and broaden its applicability to realistic training scenarios. These developments can be organized into three key themes, (1) spectral properties that determine what functions networks learn efficiently, (2) dynamics in finite-width networks that go beyond the kernel regime, and (3) applications to robustness and optimization.

2.2.1 Spectral properties and what networks learn

Cao et al. (2021) rigorously proved spectral bias through NTK eigendecomposition, showing that networks learn lower-degree spherical harmonics faster, explaining frequency-dependent learning dynamics (Fig. 4). Yu et al. (2023) extended this to non-uniform data distributions, showing that sampling density modulates effective frequency bandwidth during training. Choraria et al. (2022) analyzed polynomial network architectures, revealing that multiplicative interactions in Π-Nets can actually speed up the learning of higher frequencies, offering a way to counterbalance standard spectral bias. Geifman et al. (2020) proved connections between NTK and classical Laplace kernels for normalized data, showing they share the same reproducing kernel Hilbert space (RKHS) properties. Hu et al. (2020b) demonstrated surprising early-time linearity in training dynamics despite nonconvexity. Cagnetta et al. (2023) analyzed wide CNNs through the NTK lens, providing a quantitative characterization of how local and hierarchical structure in image data shapes the NTK spectrum and governs generalization rates, explaining why convolutional inductive bias is effective for high-dimensional spatial data.

Hacohen and Weinshall (2022) revisited the over-parameterized deep linear network model and proved that convergence rates are exponentially faster along directions of larger principal components. This establishes a principal components bias that explains why networks of different

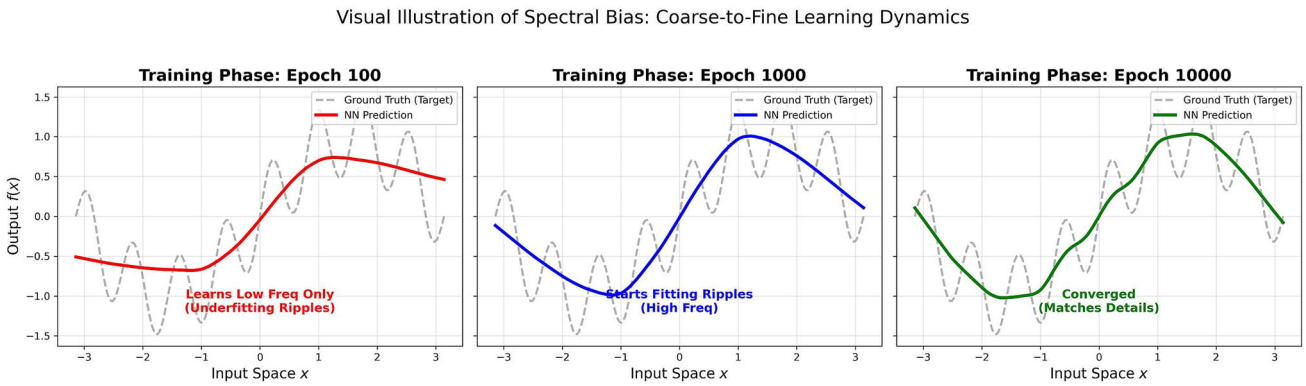


Fig. 4 Visual demonstration of spectral bias in neural network training. A neural network trained on a composite target function $f(x) = \sin(x) + 0.5\sin(8x)$. Consistent with NTK spectral analysis (Cao et al. 2021), the network learns the low-frequency component (left subplot) rapidly, effectively smoothing the data. It requires significantly more training epochs to capture the high-frequency "ripples" (middle subplot), eventually converging to the detailed function (right subplot). This demonstrates the inductive bias of neural networks towards smooth, low-complexity solutions

architectures learn to classify images in the same order. Geiger et al. (2021) provided a geometric analysis of the deep learning loss landscape through the NTK lens. They showed that the concentration of high-dimensional data near low-dimensional manifolds is what enables gradient descent to escape the curse of dimensionality. Zhang et al. (2020b) exploited NTK-regime linearity to derive an explicit and quantitative characterization of how initialization affects generalization error in regression. They proved that initialization-induced bias persists throughout training and is directly quantifiable through kernel theory.

2.2.2 Finite-width dynamics and feature learning

Moving beyond the infinite-width kernel regime reveals richer learning dynamics where networks actively learn features rather than performing kernel regression with fixed representations. Woodworth et al. (2020) characterized how initialization scale controls the transition between the lazy kernel regime and the rich regime. In the lazy regime, networks behave as kernelized linear predictors. In the rich regime, gradient descent induces feature learning beyond reproducing kernel Hilbert spaces. Their complete analysis for multilayer homogeneous models determines when NTK theory accurately predicts behavior versus when feature learning effects dominate. This lazy-to-rich transition has been connected to the grokking phenomenon by Kumar et al. (2024). They demonstrated that sudden generalization after prolonged training arises when a network first fits a kernel solution with initial features, followed by a late-time feature learning phase where a generalizing solution is identified. Damian et al. (2022) proved that gradient descent on sufficiently wide networks can learn representations that provably outperform any fixed kernel. They demonstrated a large class of functions learnable by gradient descent but not by any kernel method, directly explaining the empirical performance gap between neural networks and their NTK approximations.

Chen et al. (2020a) proved that finite-width networks learning hierarchical representations achieve exponential sample complexity advantages over kernel methods including the NTK. Specifically, they demonstrated $\tilde{O}(d^{\lceil p/2 \rceil})$ sample complexity for degree-p polynomials compared to $\tilde{O}(d^{p-1})$ for NTK-based methods. This reveals how neural networks exploit intermediate representations to achieve improved efficiency. Allen-Zhu et al. (2019) extended analysis beyond NTK linearization by establishing quadratic approximations. These explain why overparameterized networks generalize despite having sample complexity nearly independent of parameter count. Mahankali et al. (2023) provided a mean-field analysis with polynomial width, samples, and time that goes beyond NTK linearization. They proved that projected gradient flow achieves better sample complexity than kernel methods without requiring artificial modifications.

Recent exact solutions for deep linear networks (Dominé et al. 2025; Braun et al. 2022) have provided unprecedented clarity on dynamics across the full training trajectory. They reveal how initialization encoding prior knowledge fundamentally shapes the NTK spectrum and accelerates the alignment of network weights with task structure. This provides a rigorous justification for the effectiveness of transfer learning and pre-training by demonstrating how initial weights influence the speed of feature discovery.

2.2.3 Robustness and optimization

Theoretical insights have directly informed optimization strategies. Hu et al. (2020c) proved that orthogonal initialization provides a provable speedup in deep linear networks by ensuring the width required for convergence is independent of depth, a significant improvement over Gaussian initialization. This optimization benefit is further supported by Karakida and Osawa (2020), who established that natural gradient descent with approximate Fisher information achieves the same fast convergence as exact natural gradient descent in wide networks, justifying the use of efficient approximations like K-FAC. Wang et al. (2021) established the first provable acceleration guarantees for Polyak momentum in the overparameterized regime, linking the acceleration directly to the NTK's eigenvalue structure. Cao and Gu (2019) provided generalization bounds for SGD that bridge NTK theory to finite-width training, establishing bounds independent of network width.

Moving beyond optimization, Banerjee et al. (2023) introduced restricted strong convexity to provide global convergence guarantees that extend beyond the strict lazy regime, supported by their sharpened $O(\text{poly}(L)/\sqrt{m})$ bound on the Hessian spectral norm. This structural understanding is complemented by Singh et al. (2021), who proved that the Hessian's inherent low-rank structure is a fundamental driver of reliable convergence. However, feature learning is not universally beneficial. Petrini et al. (2022) demonstrated that learning sparse features can paradoxically degrade performance in fully-connected networks while remaining essential for CNNs. Bordelon and Pehlevan (2023) quantified the $O(1/\sqrt{\text{width}})$ fluctuations of the NTK, providing a rigorous framework for uncertainty estimates and prediction variance in finite-width training dynamics.

Gao et al. (2019) established NTK-based convergence guarantees for adversarially trained networks, proving robust interpolation requires significantly higher model capacity than standard interpolation. Allen-Zhu and Li (2022) proved that adversarial training purifies networks by eliminating non-robust features, providing a mechanistic explanation for how adversarial training restructures learned representations.

Loo et al. (2022) tracked NTK evolution under adversarial constraints, revealing that robust models develop distinct spectral structures, while Tsilivis and Kempe (2022) showed that robust features align with the kernel's largest eigenvalues. Fort et al. (2020) conducted a systematic study of NTK time evolution alongside loss landscape geometry, demonstrating that a highly chaotic rapid initial transient determines the final basin of low loss, during which the NTK learns useful features.

Wei et al. (2019) proved maximum normalized margin solutions emerge from cross-entropy minimization and demonstrated linear versus quadratic sample complexity gaps between feature learning and NTK approaches. Nitanda and Suzuki (2020) derived minimax optimal convergence rates for averaged SGD in the NTK regime, justifying Polyak averaging theoretically. Hu et al. (2021) studied overparameterized networks from a nonparametric perspective, establishing that ℓ_2-regularized gradient descent achieves near-optimal estimation rates for recovering smooth target functions. Ye et al. (2023) analyzed how overparameterization impacts privacy in the NTK regime, proving that wider networks near initialization have tighter privacy bounds due to the reduced sensitivity of NTK predictions to individual training points.

2.3 Applications to trustworthy system design

NTK analysis provides theoretical foundations for trustworthy knowledge-based systems through uncertainty quantification, physics-informed learning, and theoretical guarantees for data-constrained scenarios. Table 5 summarizes how NTK analysis provides interpretable understanding and drives practical improvements across these diverse applications.

He et al. (2020) proved that Bayesian deep ensembles yield analytically distinct, conservative posterior predictives via NTK-Gaussian process (NTK-GP) connections, providing a rigorous basis for uncertainty estimation. Shi et al. (2024) extended this to federated settings, demonstrating that NTK-based ensemble models provide improved uncertainty quantification under data heterogeneity. Huang et al. (2021) further quantified how data heterogeneity affects the effective NTK convergence in federated learning architectures.

In the domain of scientific computing and compression, McClenny and Braga-Neto (2023) utilized NTK spectral analysis to explain how self-adaptive physics-informed neural networks (SA-PINNs) prioritize difficult solution regions, while Arora et al. (2020) traced NTK's efficacy in low-data regimes to reduced output variance. Ji and Zhu (2020) derived NTK-based risk bounds for knowledge distillation, while Nguyen et al. (2021) explained joint autoencoder (AE) training superiority through favorable NTK spectral properties. For graph-structured data, Wang et al. (2024b) and Xu et al. (2023) developed structure-based NTKs for graph dataset distillation, proving that distilled datasets preserve the essential spectral properties of the original graph distribution. Wang et al. (2022b) established MetaNTK convergence guarantees for meta-learning, enabling a 100× speedup in neural architecture search. Wei et al. (2023) proposed NTK-approximating MLP fusion for efficient language model fine-tuning, demonstrating that NTK approximations of pre-trained blocks can be merged into compact MLPs for edge devices. Chu et al. (2021) applied the NTK framework to unsupervised 3D shape completion, proving that the NTK of an untrained network encodes a deep prior that enables high-quality reconstruction without training data.

Table 5 Summary of NTK applications across diverse domains with key insights and practical impacts

Domain	Key Insight from NTK Analysis	Practical Impact
Physics-Informed	Self-adaptive weights equalize NTK eigenvalues (McClenny and Braga-Neto 2023)	Superior partial differential equation solutions, fewer epochs
Ensemble Methods	Bayesian ensembles via NTK-GP connection (He et al. 2020)	Conservative predictions, out-of-distribution robustness
Knowledge Transfer	Distillation risk bounds via NTK (Ji and Zhu 2020)	Soft labels carry structural information
Low-Data Learning	NTK reduces output variance in wide nets (Arora et al. 2020)	1–3% gain on CIFAR-10
Meta-Learning	Model-agnostic meta-learning (MAML) convergence ≡ MetaNTK regression (Wang et al. 2022b)	100× faster neural architecture search
Federated Learning	Heterogeneity affects effective NTK convergence (Huang et al. 2021; Shi et al. 2024)	Improved aggregation and uncertainty
Graph Learning	Structure-based NTK for dataset distillation (Wang et al. 2024b; Xu et al. 2023)	Massive compression; accuracy preserved
LLM Fine-tuning	NTK-approximating MLP fusion (Wei et al. 2023)	Efficient fine-tuning on edge devices
3D Reconstruction	Deep prior encodes shape proximity in NTK space (Chu et al. 2021)	Unsupervised shape completion

2.4 Understanding architectures through NTK analysis

NTK theory reveals how architectural choices determine what networks learn and why certain designs succeed. Tachella et al. (2021) identified a mathematical link between untrained convolutional neural network (CNN) denoisers and classical non-local filtering algorithms such as BM3D, proving that denoising is an intrinsic property of the network's architecture and initialization. This allows signal processing principles to directly inform model design. Chen et al. (2021a) formalized the equivalence between infinitely wide networks trained with soft-margin loss and support vector machines, expanding NTK theory to a broader family of regularized kernel machines.

NTK theory also characterizes architectural capabilities across specialized domains. Xu et al. (2021b) utilized NTK analysis to determine the extrapolation limits of graph neural networks, proving they possess distinct generalization profiles when encountering structures unseen during training. Chen et al. (2020b) demonstrated that kernel-like behavior and linear convergence persist under practical scenarios involving noisy gradient descent and weight decay. In quantum computing, Liu et al. (2022) and Incudini et al. (2023) developed the Quantum NTK and the Quantum Path Kernel, respectively. These frameworks establish that variational quantum circuits can emulate deep learning's hierarchical advantages and converge to predictable Gaussian process behavior in the large-width limit.

Finally, NTK spectral properties have emerged as efficient proxies for architectural evaluation. Mok et al. (2022) demonstrated that NTK condition numbers at initialization can reliably predict post-training accuracy, enabling a $10\times$ speedup in neural architecture search. However, the framework also highlights that the highly non-linear evolution of modern architectures can limit the reliability of static NTK proxies. Paccolat et al. (2021) showed that while lazy networks remain static, networks in the feature-learning regime evolve to compress uninformative input directions onto a task-relevant manifold, providing a geometric explanation for how networks learn invariant representations.

2.5 Finite-width networks and practical considerations

The infinite-width limit provides theoretical tractability. However, practical interpretability requires understanding how and when finite networks deviate from this idealization. Seleznova and Kutyniok (2022a, b) mapped these boundaries extensively, revealing that NTK validity is contingent on specific architectural phases. While standard initializations often preserve kernel behavior, combining deep architectures with specific initialization variances can drive networks into a chaotic phase where the kernel remains random and evolves significantly, defying the frozen kernel assumption. Taheri and Thrampoulidis (2024) established that a minimum polylogarithmic width of $m = \Omega(\log^4(n))$ is sufficient for NTK-based stability and generalization, a much tighter bound than the polynomial width required by earlier theories. Conversely, for narrow architectures, Lee et al. (2022) extended the framework to the infinite-depth limit, demonstrating that trainability guarantees can be recovered if depth is leveraged as the primary scaling dimension. Bordelon and Pehlevan (2023) further quantified $O(1/\sqrt{\text{width}})$ fluctuations of NTK predictions, establishing the rigorous correction terms needed to bridge mean-field theory to practical network sizes.

High-dimensional NTK analysis has also resolved puzzles regarding generalization complexity. Adlam and Pennington (2020) characterized generalization error in the deep overparameterized regime, providing the theoretical basis for triple descent. Their work explains why test error can exhibit additional peaks after the interpolation threshold when the parameter count scales quadratically with the data size. Mallinar et al. (2022) proposed a spectral taxonomy for overfitting, identifying ReLU networks as having tempered overfitting, where noise absorption is mitigated by the power-law decay of NTK eigenvalues. Xu and Gu (2023) extended these findings beyond the lazy regime to non-smooth networks, proving that tempered overfitting persists even during active feature learning.

Finally, NTK theory refines our understanding of how models interact with noisy or specific training data. In the presence of corrupted labels, Hu et al. (2020a) proved that anchoring parameters to their initial state via regularization allows the network to generalize well on clean data by acting as a robust filter, yielding bounds comparable to noise-free training. Zhang and Zhang (2022) adapted classical influence functions to the overparameterized regime, demonstrating that while standard influence analysis is fragile due to Hessian instability, NTK-based modifications can rigorously quantify the impact of individual training examples on final model predictions.

2.6 Recent developments and emerging directions

Recent advancements have pushed the boundaries of NTK theory into more complex architectures and data regimes. Novak et al. (2022) developed fast finite-width NTK computation algorithms that exploit architectural sparsity, enabling million-parameter network interpretability. Han et al. (2024) extended the framework to diffusion models, establishing early sample complexity guarantees for score matching. Xu and Zhu (2024) established uniform SGD concentration results for multi-layer networks under realistic width and streaming data constraints, bridging theory and practice.

New theoretical paradigms are also emerging to unify kernel methods with active feature learning. Yang et al. (2023) proposed a theory of representation learning that generalizes kernel methods by parameterizing over data-dependent feature maps, spanning both lazy and rich regimes within a unified framework. Akiyama and Suzuki (2023) proved that two-layer ReLU networks achieve strictly lower excess risk than any kernel method, providing a quantitative measure of the performance gap induced by non-convex feature learning.

Furthermore, the limits of the kernel perspective have been more clearly defined. Allen-Zhu and Li (2023) proved that deep networks learn features enabling knowledge transfer and ensembling benefits that are impossible in the frozen kernel regime. Kothapalli and Tirer (2025) and Seleznova et al. (2023) bridged the NTK and neural collapse phenomena, proving that the NTK's emerging block eigenstructure serves as a spectral signature of collapsed representations. Vyas et al. (2023) demonstrated empirically that NTK proxies fail to predict the scaling laws of real neural networks, establishing concrete boundaries for kernel-based interpretability. Shi et al. (2023) proposed a unified gradient feature learning framework that recovers NTK predictions in the lazy limit while providing stronger guarantees for feature discovery in practical settings.

2.7 Critical analysis

The NTK framework provides powerful theoretical insights with several important limitations. The infinite-width lazy regime, while mathematically tractable, increasingly diverges from practice: modern networks operate with small initialization in the rich regime where the kernel evolves significantly and theoretical guarantees break down. Bartlett et al. (2021) showed that NTK theory systematically underestimates generalization in the feature learning regime, while Vyas et al. (2023) demonstrated that NTK proxies fail to capture scaling law behavior. Together, these findings establish that the kernel lens provides reliable interpretability near initialization but offers only partial insight into full training trajectories. The lazy-to-rich transition (Woodworth et al. 2020) and exact deep linear network solutions (Dominé et al. 2025) provide partial bridges, but a complete nonlinear theory of the rich regime remains elusive.

Computationally, despite algorithmic advances (Novak et al. 2022), analyzing billion-parameter networks through the kernel lens remains prohibitive. Moreover, the NTK itself can be as complex as the network it describes, raising questions about whether kernel matrix analysis genuinely simplifies interpretation for practitioners. These limitations do not negate the framework's value because a partial but rigorous understanding is more valuable than empirical intuition. However, they establish a clear boundary. The NTK is a reliable theoretical tool for understanding behavior near initialization, comparing architectures, and analyzing training dynamics, but it provides incomplete insight into the representation learning that drives practical deep learning success.

2.8 Summary

This section reviewed 82 seminal and recent works connecting the NTK framework to DNN interpretability. The NTK enables rigorous analysis of training dynamics, generalization, and architectural bias by connecting neural networks to Gaussian processes and support vector machines (SVMs). Key theoretical developments clarify that lazy-regime networks behave as kernel machines, while practical networks operate in the rich regime where feature learning dominates. New theoretical results establish connections between NTK spectral properties and principal components bias (Hacohen and Weinshall 2022), nonparametric estimation optimality (Hu et al. 2021), privacy guarantees (Ye et al. 2023), finite-width fluctuations (Bordelon and Pehlevan 2023), and neural collapse (Seleznova et al. 2023). Applications span SA-PINNs, ensemble methods, federated learning, graph distillation, LLM fine-tuning, and 3D reconstruction. Future work must bridge static kernel theory and dynamic feature learning toward a unified theory of representation learning.

In terms of applicability, the NTK framework is currently best suited to comparing architectures, analyzing training dynamics, and characterizing convergence and generalization in the over-parameterized regime. This applies particularly to wide fully-connected and convolutional networks near initialization, rather than to finite-width networks operating in the rich feature-learning regime.

3 Sparse representation methods

Sparse representation theory provides a complementary framework for understanding deep learning through the lens of efficient coding principles. While the NTK framework establishes connections between neural networks and kernel methods, sparse representation theory reveals how neural networks learn to encode information using few active elements from a larger set of possible features. This framework has proven particularly powerful for constructing interpretable neural network architectures through algorithm unrolling. In this approach, classical iterative sparse coding algorithms are transformed into deep networks with clear physical interpretations.

Unlike dense representations that activate most network units, sparse representations activate only a few. This provides advantages in computational efficiency, interpretability, and biological plausibility (Table 6). These properties make sparse coding particularly attractive for constructing interpretable neural network architectures.

Table 6 Comparison of dense and sparse representations in neural networks. Sparse representations provide computational efficiency, interpretability, and biological plausibility at the cost of more complex optimization

Property	Dense Representation	Sparse Representation
Activation Pattern	Most units active	Few units active
Information Encoding	Distributed across all units	Concentrated in active units
Computational Cost	$O(n)$ multiplications	$O(k)$ where $k \ll n$
Memory Efficiency	Full storage required	Compressed storage possible
Interpretability	Entangled features	Disentangled, localized features
Biological Plausibility	Less aligned with cortex	Consistent with neural coding
Optimization	Standard gradient descent	Requires ℓ_1 penalties or thresholding
Robustness	Sensitive to noise	More robust to noise

3.1 Theoretical foundations

Olshausen and Field (1996) established the sparse coding framework, showing that sparse representations naturally produce interpretable, localized basis functions. This work established sparsity as a fundamental principle, replacing earlier independence assumptions in signal processing. Olshausen et al. extended this framework in Olshausen and Field (1997) to the overcomplete case where the number of basis functions exceeds the input dimensionality ($m > n$). In overcomplete representations, the basis functions are non-orthogonal and linearly dependent, making the sparse coding problem more challenging but also more powerful. Sparsification in the overcomplete setting recruits only those basis functions necessary for representing the input, enabling more flexible and efficient representations.

The sparse coding model represents an input signal $\mathbf{x} \in \mathbb{R}^n$ as a linear combination of dictionary elements (basis functions) with sparse coefficients as

$$\mathbf{x} = \mathbf{D}\mathbf{z} + \boldsymbol{\epsilon} \tag{3}$$

where $\mathbf{D} \in \mathbb{R}^{n \times m}$ is the dictionary matrix whose columns are the basis functions, $\mathbf{z} \in \mathbb{R}^m$ is the sparse code vector with most entries being zero, and $\boldsymbol{\epsilon}$ represents reconstruction error. The key constraint is sparsity, which requires that only a small fraction of the code elements should be active for any given input. This is formalized through the optimization problem

$$\min_{\mathbf{z}} \frac{1}{2}\|\mathbf{x} - \mathbf{D}\mathbf{z}\|_2^2 + \lambda\|\mathbf{z}\|_1 \tag{4}$$

known as the LASSO or basis pursuit denoising, where the ℓ_1 penalty promotes sparsity in the solution. The parameter λ controls the trade-off between reconstruction fidelity and sparsity. This overcomplete sparse coding model motivated dictionary learning algorithms such as K-SVD and online dictionary learning (Aharon et al. 2006; Mairal et al. 2009), which alternatively optimize both dictionaries and sparse codes for efficient signal representation.

The connection between sparse coding and neural networks was formalized through the iterative shrinkage-thresholding algorithm (ISTA) (Daubechies et al. 2004), which solves the LASSO problem (Tibshirani 1996) through iterative updates

$$\mathbf{z}_{(k+1)} = \mathcal{S}_{\lambda/L}\left(\mathbf{z}_{(k)} + \frac{1}{L}\mathbf{D}^T(\mathbf{x} - \mathbf{D}\mathbf{z}_{(k)})\right) \tag{5}$$

where $\mathcal{S}_\theta$ is the soft-thresholding (shrinkage) operator defined as $\mathcal{S}_\theta(x) = \text{sign}(x)\max(|x| - \theta, 0)$, and L is a step size parameter related to the largest eigenvalue of $\mathbf{D}^T\mathbf{D}$. This iterative structure directly maps to recurrent neural network computations, where each iteration corresponds to a layer of processing with linear transformations followed by nonlinear activation, as illustrated in Fig. 5.

The paradigm of algorithm unrolling, introduced by Gregor and Lecun (2010), transforms these classical iterative optimization algorithms into feedforward neural networks. Their learned ISTA (LISTA) network unrolls ISTA iterations into network layers with learnable parameters, enabling the network to learn optimal thresholding schedules and weight matrices through backpropagation. This typically requires far fewer layers than ISTA iterations while maintaining interpretability. It should be noted that ISTA solves for sparse codes column-wise, processing each input sample independently. Alternative approaches such as consistent adaptive sequential dictionary (CASD) (Seghouane and Iqbal 2018) update each row of the coefficient matrix at a time, offering computational advantages by processing all samples simultaneously for each dictionary atom. However, deep unfolding networks predominantly adopt the column-wise formulation due to its natural correspondence with feedforward architectures.

Recent theoretical work has rigorously formalized the connection between sparse coding and deep learning. Papyan et al. (2017) established that CNNs with ReLU activations

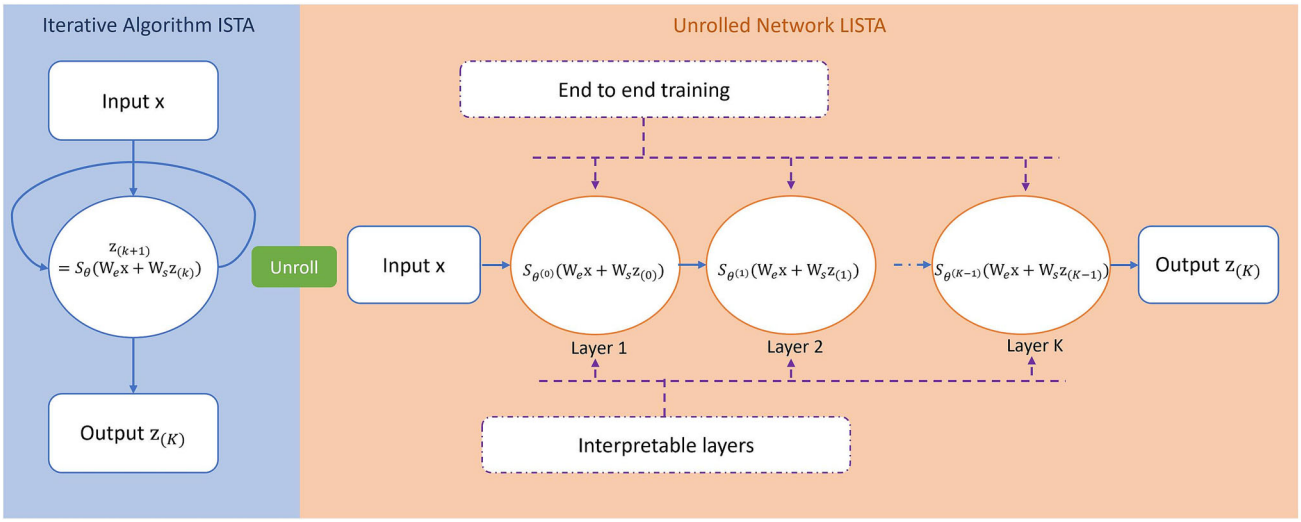


Fig. 5 Algorithm unrolling transforms iterative sparse coding algorithms into deep neural networks. Left subplot is the ISTA algorithm iteratively applying a fixed soft-thresholding operator $\mathcal{S}_\theta$ to solve the sparse coding problem. Right is the unrolled network (LISTA) mapping each iteration to a distinct layer, where the single fixed threshold becomes layer-specific learnable parameters $\theta^{(0)}, \theta^{(1)}, \ldots, \theta^{(K-1)}$. This allows the network to learn an optimal thresholding schedule, typically more aggressive sparsification in early layers followed by refinement, while maintaining interpretability through end-to-end training

can be interpreted as pursuit algorithms for hierarchical sparse coding. The CNN forward pass implements the Layered Thresholding pursuit, where ReLU activations act as soft non-negative thresholding operators, filters correspond to convolutional dictionaries, and bias terms function as sparsity-controlling regularization parameters. Their analysis provides theoretical guarantees for the uniqueness and stable recovery of layer-wise features under local $\ell_{0,\infty}$ sparsity conditions. Building on this foundation, Sulam et al. (2018) developed projection-based pursuit algorithms for the multi-layer convolutional sparse coding (ML-CSC) model with enhanced stability guarantees, demonstrating that CNN computations estimate hierarchically nested sparse representations. Papyan et al. (2018) further consolidated these foundations, showing that alternatives to the standard forward pass such as Layered Basis Pursuit and Layered ISTA offer superior theoretical stability by iterating activations to better satisfy the underlying sparse model. Extending ML-CSC interpretations beyond feedforward architectures, Zhang and Zhang (2021) constructed mathematically equivalent formulations for residual and dilated dense networks. Their analysis proves that ResNet shortcut connections provide superior initialization for sparse pursuit at each layer, addressing the degradation phenomenon, while dense connections in MSDNet correspond to augmenting dictionaries with identity matrices that reduce mutual coherence $\mu(\mathbf{D})$ and improve recovery performance. These advances collectively establish that modern deep architectures implicitly implement classical sparse coding principles, providing a rigorous mathematical foundation for their interpretability.

3.2 Algorithm unrolling: bridging sparse coding and deep learning

In LISTA, the iterative update in (5) is recast as

$$\mathbf{z}_{(k+1)} = \mathcal{S}_{\boldsymbol{\theta}^{(k)}}\left(\mathbf{W}_e\mathbf{x} + \mathbf{W}_s\mathbf{z}_{(k)}\right) \tag{6}$$

where $\mathbf{W}_e$ and $\mathbf{W}_s$ are learnable weight matrices that can differ across layers, and $\boldsymbol{\theta}^{(k)}$ are learnable thresholds. In the original ISTA these matrices are constrained ($\mathbf{W}_e = \frac{1}{L}\mathbf{D}^T$ and $\mathbf{W}_s = \mathbf{I} - \frac{1}{L}\mathbf{D}^T\mathbf{D}$), however, LISTA relaxes these constraints and learns optimal values through training. This allows the network to achieve the same sparse coding quality as many ISTA iterations using far fewer layers, typically 10-20 layers suffice where ISTA might require hundreds of iterations.

The algorithm unrolling framework has since been extensively developed and formalized. Monga et al. (2021) provided a comprehensive survey of algorithm unrolling as a network design methodology, systematically reviewing how to unfold iterative algorithms into layer-wise neural architectures for signal and image processing tasks. As synthesized in their review, unrolling-based architectures offer three key properties, (1) *interpretability*, where every network component corresponds to specific operations in the underlying optimization algorithm, (2) *efficiency*, where learned parameters enable faster convergence than fixed algorithmic counterparts, and (3) *performance*, where the flexibility of learnable parameters typically improves results over the original algorithm.

Theoretical understanding of LISTA has been advanced by several key works. Chen et al. (2018) proved that LISTA achieves linear convergence rates, providing theoretical guarantees that match empirical observations. Their study explicitly mapped the necessary evolution of weights and thresholds across layers, proving that a specific coupled weight structure is required to ensure optimal convergence. This theoretical grounding sets unrolled models apart from standard black-box deep learning, allowing practitioners to reason about network behavior using classical optimization theory. Schnoor et al. (2023) extended these theoretical guarantees beyond convergence rates, deriving generalization error bounds for a broad class of unrolled networks ranging from recurrent to feedforward architectures. Their work established that sample complexity essentially depends linearly on the number of parameters and on the depth, proving that greater weight sharing yields tighter out-of-sample recovery guarantees. Moreau and Bruna (2017) provided a deeper geometric explanation for this acceleration. Through matrix factorization analysis, they proved that LISTA implicitly learns an optimal preconditioning of the sparse coding problem. The learned weight matrices effectively diagonalize the dictionary Gram matrix $\mathbf{D}^T\mathbf{D}$, transforming an ill-conditioned optimization landscape into a favorable, well-conditioned one. This reveals that LISTA's acceleration is geometrically principled rather than heuristic, establishing a rigorous mathematical bridge between algorithm unrolling and the matrix factorization theory reviewed in Section 4.

Dai et al. (2022) demonstrated that sparse convolutional models such as SDNet achieve classification performance comparable to conventional deep networks while offering enhanced interpretability and robustness. By replacing standard convolutional operations with optimization-based CSC-layers, they leveraged the stable recovery properties of sparse modeling to provide superior robustness against input corruptions and adversarial perturbations. Crucially, this robustness is tunable through regularization strength and the number of pursuit iterations, with even 2–3 FISTA iterations sufficient for high accuracy, bridging the gap between principled sparse models and high-performance deep learning. Early application of algorithm unrolling to non-negative constraints was demonstrated by Wisdom et al. (2017), who introduced Deep Recurrent NMF (DR-NMF) for speech source separation. By unrolling a sequential ISTA algorithm designed for sparse NMF of audio spectrograms, DR-NMF inherits principled initializations from NMF model parameters, maintaining direct correspondence between network weights and physical audio components while enabling faster convergence and greater interpretability than standard recurrent architectures.

3.3 Connections between deep networks and sparse coding

The relationship between CNNs and sparse coding has been formally established through several theoretical frameworks. Papyan et al. (2017) demonstrated that CNNs with ReLU activations implement iterative sparse recovery for hierarchical representations, with each forward pass computing approximate solutions to nested CSC problems. Their ML-CSC framework guarantees unique and stable recovery of layer-wise features under local sparsity assumptions. Sulam et al. (2018) developed projection-based pursuit algorithms with improved stability bounds, while their subsequent work (Sulam et al. 2020) generalized this to global optimization frameworks. Unlike layer-by-layer approximations, their multi-Layer ISTA (ML-ISTA) shows that recurrent neural networks (RNNs) naturally implement end-to-end pursuit algorithms that jointly optimize sparse codes across all depths with proven convergence properties.

Tolooshams et al. (2021) established a formal link between deep autoencoders and classical iterative algorithms through an EM-inspired framework. Their constrained recurrent sparse AE (CRsAE) interprets dictionary learning as an approximate EM algorithm. The encoder (E-step) solves a sparse coding problem via FISTA, while the decoder (M-step) updates the dictionary. This architecture enables the simultaneous training of the dictionary and regularization parameters, providing rigorous theoretical grounding for AE-based dictionary learning. The stability of these representations was further characterized by Romano et al. (2020), who proved that adversarial robustness is directly quantifiable through signal sparsity and dictionary coherence. By modeling signals within the ML-CSC framework, they demonstrated that robustness is tied to the local geometry of sparse signal manifolds, offering a principled explanation for why sparse-structured representations are inherently more stable than dense alternatives.

Complementing these optimization perspectives, Giryes et al. (2016) investigated geometric properties using compressed sensing theory, establishing that networks with random Gaussian weights act as near-isometric embeddings. This preserves pairwise distances and provides differential treatment for within-class versus across-class relationships, offering theoretical support for the effectiveness of random initialization.

3.4 Deep unfolding for interpretable architectures

The algorithm unrolling paradigm has been extensively applied to construct interpretable deep architectures, bridging optimization theory with data-driven learning. For image

restoration, unfolding proximal algorithms yields superior generalization through distinct role separation. Wang et al. (2020a) developed RCDNet, where learned rain kernels explicitly characterize streak patterns and proximal operators implement interpretable denoising. Li et al. (2020) tackled blind deblurring through deep unrolling for blind deblurring (DUBLID), unrolling half-quadratic splitting to transparently separate blur estimation from reconstruction. Cao et al. (2022) unified CSC with traditional detail injection for pansharpening.

For multi-modal applications, where interpretability is crucial for understanding cross-modal fusion, sparse coding principles effectively regularize heterogeneous data integration. Deng et al. (2024) introduced DeepM^{2}CDL for multi-scale hierarchical fusion, while Marivani et al. (2020) incorporated sparse priors for super-resolution. Deng et al. (2023) formalized disentangled CSC (DCSC) for image registration, explicitly separating alignment features from modality-specific characteristics. For constrained optimization, Huang and Dragotti (2022) proposed WINNet, achieving perfect reconstruction through wavelet-inspired invertible transformations.

3.5 Applications in specific domains

Algorithm unrolling provides interpretability across diverse domains by transforming opaque neural computations into transparent optimization steps where each layer's function is mathematically defined and visually inspectable. The following applications demonstrate how this interpretability-by-design paradigm enables trustworthy deployment in safety-critical systems.

3.5.1 Compressive sensing and image reconstruction

Compressive sensing and artifact removal represent natural application domains for sparse coding-based networks. For JPEG artifact reduction, Fu et al. have pioneered model-driven approaches. Their earlier work (Fu et al. 2019) utilized dilated convolutions within a learned iterative shrinkage-thresholding framework to handle multiple compression levels. Their subsequent work (Fu et al. 2022) explicitly unfolded a maximum a posteriori (MAP) model, combining data-driven flexibility with model-driven interpretability.

Recent advances have moved toward multi-scale and dual-domain optimization to handle complex image degradations. Li et al. (2024) developed D3C2-Net, which integrates image-domain and coding-domain priors through a dual-pathway architecture that exploits sparse priors in both domains simultaneously. Addressing the limitations of single-scale models, Xu et al. (2022) proposed multi-scale convolutional sparse coding network (MCSCNet), which unrolls a multi-scale CSC model to extract features across multiple resolutions, significantly improving denoising performance while maintaining a lightweight parameter footprint. Adaptive parameterization has also proven critical for generalization. Janjušević et al. (2022) introduced CDL-Net, demonstrating that noise-adaptive thresholds enable robust performance across varying noise levels. They further expanded this in GroupCDL (Janjušević et al. 2025) by incorporating learned group-sparsity and circulant attention to model nonlocal self-similarity, achieving parity with black-box models in compressed sensing magnetic resonance imaging (MRI) reconstruction. Similarly, Wang et al. (2024a) extended RCDNet with dynamic kernel inference to adapt to diverse rain streak characteristics. This proves that input-dependent parameterization allows unrolled networks to generalize effectively across varied degradation scenarios.

In the medical imaging domain, Wang et al. (2022a) proposed DICDNet, a deep interpretable convolutional dictionary network for metal artifact reduction (MAR) in CT images. Unlike generic restoration modules, DICDNet explicitly models metal artifacts as sparse combinations of learned dictionary atoms, directly embedding the physical artifact formation process into the network structure. This design ensures that learned basis functions correspond to interpretable streaking and star-shaped artifact patterns, providing both high reconstruction fidelity and clinically meaningful transparency. For natural image denoising, Yin and Wang (2023) introduced SGSC-Net, which develops a side group sparse coding model exploiting group sparse coefficient consistency as an explicit prior. By unrolling this structured sparsity model into a deep network, SGSC-Net provides interpretable parameters tied to group sparsity regularization, where each layer's thresholding operation has a clear mathematical meaning rooted in the underlying prior model.

3.5.2 Video, graph, and temporal signal processing

Extending unrolling beyond static images, Luong et al. (2021) presented interpretable RNNs derived from reweighted ℓ_1 minimization. By mapping the iterative optimization of temporal signals to recurrent units, their architecture incorporates temporal sparse priors, demonstrating enhanced representational capacity and faster convergence than conventional RNNs. In the non-Euclidean domain, Chen et al. (2021b) introduced Graph Unrolling Networks (GUN), which employ learnable kernel functions to determine edge-specific weights. Their approach, specifically Graph Unrolling Sparse Coding (GUSC), achieved significantly lower denoising error than traditional graph Laplacian and wavelet methods by adaptively capturing signal priors. Extending the unrolling paradigm to graph neural network (GNN) interpretation, Miao et al. (2023) presented a unified optimization framework demonstrating that the message-

passing aggregation steps of diverse GNN models including graph convolutional network (GCN), GraphSAGE, and graph attention network (GAT) can all be derived as optimal solutions to distinct sparse regularized graph signal reconstruction problems. Their hybrid regularized framework not only provides interpretable explanations for established GNN designs but also guides the development of new architectures such as GNN-MD, which utilizes theoretically justified aggregation mechanisms based on model-driven and data-driven dictionary learning.

3.5.3 Mechanical and biomedical signal processing

In safety-critical industrial applications, interpretability is a requirement rather than a luxury. The sparse unrolling paradigm has produced a substantial family of transparent fault diagnosis architectures that directly address this requirement. Zhao et al. (2022) developed LGSC-Net, a model-driven deep unrolling method motivated by multi-layer sparse coding and explicitly targeting adversarial noise robustness. Their ante-hoc interpretable network unrolls a general sparse coding algorithm, inheriting robustness properties linked to the sparsity level of learned representations. Physical wave-propagation models have motivated a distinct family of interpretable architectures for structural health monitoring. Zhang et al. (2022) proposed an interpretable Lamb wave CSC method (LW-CSC) that embeds dispersive wave physics directly into the dictionary design. By constraining basis functions to toneburst signals at physically meaningful center frequencies, the network allows practitioners to trace decisions back to identifiable physical wave modes, enabling damage identification and localization with full physical transparency.

Across the fault-diagnosis family that follows, the unrolled algorithm is the primary axis of variation. Successive works swap the underlying optimizer (ISTA, alternating direction method of multipliers (ADMM)) or the dictionary structure (single versus multi-layer), while preserving a single interpretability mechanism, namely a one-to-one correspondence between network layers and sparse-coding iterations. The methods below should therefore be read as design choices along this axis rather than as independent architectures.

For classical ISTA-based unrolling, An et al. (2022) proposed NISTA-Net to visualize learned fault features, while Rao et al. (2024) enhanced this with CISTA-Net, adding parallel reconstruction modules to verify feature validity. Pushing this further, An et al. (2024) integrated unrolling into an adversarial framework (AAU-Net), combining mechanism-driven encoders with generative capabilities for mechanical anomaly detection. Qin et al. (2024) developed a multi-layer convolutional dictionary learning (ML-CDL) network by unrolling ML-CLISTA hierarchically, capturing vibration features across multiple frequency scales in a single interpretable architecture with stability guarantees inherited from ML-CSC theory. Yuan et al. (2025) introduced GADMM-Net, which unrolls the ADMM for general CSC. The ADMM formulation naturally separates sparse feature extraction from signal reconstruction, making explicit the distinct roles of sparsification and reconstruction in the diagnostic pipeline with enhanced convergence relative to ISTA-based approaches. In the biomedical sphere, Hou et al. (2023) applied Half-Quadratic Splitting (HQS) to electrocardiogram (ECG) denoising, using correlation-derived adaptive hyperparameters to unify traditional signal processing with neural architectures. Song et al. (2021) demonstrated that sparse coding principles extend to digital pathology through sparse coding driven deep decision tree ensembles (ScD2TE) for nucleus segmentation.

Their hybrid approach maintains interpretability at both the feature extraction level through dictionary atoms and the decision level through tree-structured classification, bridging sparse coding with symbolic reasoning in biomedical image analysis. Taken together, these architectures show that the interpretability of unrolled sparse models is robust to the choice of optimizer and dictionary depth. In each case the diagnostic decision traces back to identifiable sparse or physical features, and the principal trade-off is between convergence speed (favoring ADMM-based unrollings) and architectural simplicity (favoring ISTA-based ones).

Sparse and dictionary-learning models have a longer history in biomedical signal analysis, including functional neuroimaging, where sparse dictionaries decompose fMRI BOLD signals into interpretable spatial and temporal components corresponding to functional brain networks (Khalid and Seghouane 2014a; Khalid et al. 2015). This established use of interpretable-by-design sparse coding in a biomedical domain parallels its more recent role in the unrolled architectures discussed above.

3.5.4 Specialized vision and learning tasks

Sparse coding principles have also enabled interpretable solutions for geometry and unsupervised learning. Kong and Lucey (2019) formulated 3D reconstruction as a hierarchical block sparse dictionary learning problem, creating a network (deep NRSfM) addressing scale and complexity challenges beyond the reach of previous methods. Bridging optimization with memory architectures, Zhou et al. (2018) reformulated adaptive ISTA as a recurrent unit connected to LSTMs (SC2Net). In the multi-view learning domain, Du et al. (2021) demonstrated that bi-sparse co-clustering optimization could be transformed into a differentiable deep network, extending sparse interpretability to unsupervised clustering tasks.

3.5.5 Hyperspectral image processing

HSI, with its extreme dimensionality, benefits significantly from the structured constraints of sparse coding. The works below differ mainly in which structural prior they add to the unrolled backbone to cope with this dimensionality, ranging from low-rank and multi-scale priors to active dictionary learning and non-negativity. Despite this variation, they share the goal of matching the sparse model to the structure of the task at hand while preserving layer-operation transparency.

For change detection, recent works have combined unrolling with low-rank constraints (Zhou et al. 2025) and semi-supervised multi-scale priors (Qu et al. 2024) to manage data scarcity and redundancy. Addressing the computational complexity of high-dimensional data, Yin and Chen (2024) introduced separable content-dependent 3D convolutions. Wu et al. (2023) addressed a critical limitation of model-driven approaches by proposing ADL-Net, a deep unrolling network with active dictionary learning for hyperspectral anomaly detection. Unlike prior models such as LRR-Net (Li et al. 2023), which separate dictionary construction from detection, ADL-Net integrates concept factorization theory into a unified end-to-end framework based on low-rank representation. The dictionary is treated as a linear combination of the original data with an orthogonal coefficient matrix, allowing it to be actively updated during detection. This eliminates the mismatch between pre-constructed dictionaries and target detection statistics while preserving mathematical transparency through layer-operation correspondence.

Xiong et al. (2021) proposed NMF-SAE, combining NMF's physical interpretability with sparse autoencoders for hyperspectral unmixing, where encoder non-negativity and sparsity constraints directly correspond to abundance estimation while the decoder provides physically meaningful endmember signatures. Theoretically, Gkillas et al. (2023) provided a unifying perspective by establishing formal connections between Deep Equilibrium Models and sparse representation, demonstrating that equilibrium-based solvers can be interpreted through the lens of iteratively unrolled sparse coding frameworks.

As summarized in Table 7, the common thread across these diverse applications is the transformation of abstract optimization steps into concrete network modules, ensuring that every layer possesses a clear physical interpretation. This interpretable-by-design approach consistently demonstrates performance matching or approaching black-box alternatives while maintaining the transparency essential for deployment in safety-critical applications.

3.6 Sparse representations for understanding neural networks

Beyond constructing interpretable architectures from scratch, sparse coding principles have recently emerged as a powerful lens for interpreting the internal representations of pre-trained black-box models. This direction has gained particular traction in analyzing large language models (LLMs) and vision Transformers.

Li et al. (2025) investigated the geometry of concepts learned by sparse autoencoders in LLMs, revealing that semantic concepts cluster naturally in the learned sparse space. This suggests that sparse coding can effectively decompose the distributed, entangled representations of massive networks into localized, meaningful features. Building on this, Bai and Ma (2025) utilized sparse representations to identify neuron-level responses to specific input features, enabling fine-grained white-box analysis of language models. Demonstrating the broader applicability of sparse decomposition methods, Khalid et al. (2025) applied dictionary learning algorithms to semantic embeddings from sentence transformers for mental health text analysis. Their CCMH system learned interpretable semantic atoms that captured clinically meaningful discourse patterns across mental health conditions. In the visual domain, Padalkar and Gupta (2025) leveraged attention-guided sparse representations to extract symbolic rules from vision Transformers, bridging the gap between sub-symbolic neural processing and human-readable logic. These developments highlight the dual utility of sparse coding. It serves both as a blueprint for designing new interpretable architectures and as a diagnostic tool for understanding existing ones.

At the intersection of sparse coding and manifold learning, Tasissa et al. (2023) proposed K-Deep Simplex (KDS), which learns a dictionary of synthetic landmark points alongside simplex-constrained representation coefficients. Their interpretable AE, designed through algorithm unrolling of an alternating minimization procedure, provably recovers the Delaunay triangulation structure of data manifolds under a local weighted ℓ_1 penalty. By establishing the equivalence between weighted ℓ_1 and weighted ℓ_0 programs under triangulation assumptions, KDS demonstrates that sparse coding provides a computationally tractable proxy for manifold topology identification, directly bridging sparse representation theory with the geometric manifold learning approaches reviewed in Section 5. Ratzon et al. (2024) examined representational drift—the spontaneous change in neural tuning observed in networks during constant environments—through the lens of implicit sparse regularization. In a navigational task, they observed that networks rapidly achieve peak performance, followed by slow sparsification of activity patterns occurring orders of magnitude more slowly than the initial learning phase. This temporal separation reveals that implicit regularization continues to operate within the zero-loss solution space, progressively simplifying representations well beyond the point of performance convergence, providing a principled explanation for how biological and artificial systems

Table 7 Representative algorithm unrolling applications across domains

Application	Base Algorithm	Key Innovation	Interpretability Benefit
JPEG Artifacts	ISTA (Fu et al.)	Multi-scale unfolding	Model-driven interpretability
Compressive Sensing	ISTA/ADMM (Li et al.)	Dual-domain optimization	Image and coding domain priors
MRI Reconstruction	Group-sparse CDL (Janjusevic et al.)	Nonlocal self-similarity prior	Parity with black-box nonlocal models
Video Reconstruction	Reweighted ℓ_1 (Luong et al.)	Temporal sparse modeling	Interpretable RNN structure
Graph Denoising	Graph sparse coding (Chen et al.)	Edge-weight-sharing conv.	Interpretable graph priors
GNN Interpretation	Hybrid regularized GSR (Miao et al.)	Unified GNN optimization framework	Derives GCN/GAT/GraphSAGE from sparse priors
Fault Diagnosis	NISTA/CISTA/ADMM (An, Rao, Yuan et al.)	General sparse coding unrolling	Visualizable fault features, tunable robustness
Structural Health	Lamb wave CSC (Zhang et al.)	Physics-informed dictionary design	Wave-mode-level physical transparency
Speech Separation	Sparse NMF/ISTA (Wisdom et al.)	Non-negative unrolled RNN	NMF parameter correspondence
3D Reconstruction	Block sparse DL (Kong et al.)	Hierarchical dictionary	Geometric interpretability
Hyperspectral Detection	LRR + active DL (Wu et al.)	Active dictionary learning	Eliminates dictionary-detection mismatch
Hyperspectral Processing	Low-rank + sparse (Zhou et al.)	Low-rank and sparse decomposition	Spectral-spatial separation
CT Artifact Reduction	Convolutional DL (Wang et al.)	Task-specific artifact dictionary	Streaking pattern-level transparency
Hyperspectral Unmixing	NMF-SAE (Xiong et al.)	NMF-constrained sparse AE	Physically meaningful endmember signatures

maintain and refine internal models during continual learning.

Wang et al. (2020b) demonstrated that domain knowledge can impose structured sparsity directly into network architecture, improving both interpretability and predictive power. Their Alternative-Specific Utility DNN (ASU-DNN) restricts each output's computation to its own input attributes, creating a sparse connectivity pattern grounded in behavioral economics. This architectural sparsity acts as a domain-knowledge-based regularizer that outperforms generic explicit and implicit regularization methods. By reducing estimation error through a lighter, more principled architecture, ASU-DNN demonstrates that sparsity derived from utility theory provides interpretable representations that mirror human decision-making patterns. These developments highlight the dual utility of sparse coding: it serves as both a blueprint for designing interpretable architectures in economics and a diagnostic tool for extracting semantic features from complex vision and language models.

Beyond economics and language models, sparse theory has also shed light on newer neural architectures. Yüce et al. (2022) showed that most implicit neural representation (INR) families including SIREN and Fourier feature networks are mathematically analogous to structured sparse dictionaries, proving this structure enables exponentially more compact signal representations and providing an interpretable lens for understanding INR architecture choices. Zhou and Ge (2023) proved that implicit regularization from gradient descent leads to benign overfitting for sparse linear regression, tightening the theoretical connection between sparse coding, implicit regularization (Section 4), and generalization in overparameterized regimes.

3.7 Critical analysis

Although sparse representation methods provide powerful frameworks for interpretable deep learning, several limitations merit careful consideration.

Computational efficiency Although LISTA reduces iterations compared to ISTA, a trade-off remains between unrolling depth and inference latency. Recent work by Xu et al. (2022) addresses this via MCSCNet, a multi-scale unrolled model that achieves state-of-the-art results with fewer parameters and lower FLOPs than standard deep learning baselines. This suggests that structured unrolling need not sacrifice computational economy, though achieving optimal efficiency–interpretability trade-offs remains an active area of research.

Approximation gaps Unrolled networks provide approximate solutions to underlying optimization problems. Despite being generally accurate within the training distribution, significant gaps can emerge between network output and theoretical sparse codes when facing out-of-distribution inputs, challenging the assumption of faithful optimization. The adversarial stability analysis of Romano et al. (2020) provides a partial theoretical characterization of these gaps, but a complete theory remains elusive.

Optimization complexity Joint optimization of dictionaries and sparse codes presents non-convex challenges that backpropagation may not fully resolve. Poorly learned dictionaries lead to suboptimal sparse codes, negating interpretability benefits. Unlike standard CNNs where feature redundancy is tolerated, sparse networks require pure learned basis functions.

Sparsity-performance trade-offs Strong sparsity constraints can limit representational capacity. While sparsity aids interpretability by disentangling features, it may discard subtle, distributed information that dense networks exploit for marginal gains. Tuning the regularization parameter λ becomes a balance between interpretability and accuracy.

Scalability to modern architectures Algorithm unrolling has been influential in signal processing; however, extending these principles to transformers and LLMs remains challenging. Attention mechanisms do not map naturally to standard sparse coding formulations as convolution does. A complementary line of work instead sparsifies attention distributions directly. Sparse-softmax formulations such as sparsemax (Martins and Astudillo 2016) and α-entmax (Correia et al. 2019) can assign exact zero weight to low-scoring tokens, producing individually sparse attention heads without post-hoc manipulation, while more recent work predicts these sparsity patterns before full attention computation (Treviso et al. 2022). Such formulations show that sparsity remains a useful structural prior for attention-based models, though they do not yet connect to algorithm-unrolling theory as convolutional sparse coding does. The emergence of sparse autoencoders for language-model interpretation (Li et al. 2025; Bai and Ma 2025) is a related but distinct direction that sparsifies internal activations rather than attention distributions. For example, InterPLM applies sparse autoencoders to protein language model embeddings to recover interpretable biological features (Simon and Zou 2025). For these SAE-based approaches, the theoretical connections to the unrolling paradigm remain underdeveloped.

3.8 Summary

This section reviewed 58 papers connecting sparse representation to DNN interpretability, highlighting the pivotal

role of sparse representation theory in bridging rigorous signal processing with modern deep learning. Starting from Olshausen and Field's foundational work establishing biological plausibility of sparse coding in the visual cortex, the field evolved toward the algorithm unrolling paradigm introduced by Gregor and LeCun. This methodology transforms iterative optimization algorithms into efficient, learnable neural networks, ensuring every layer possesses clear physical interpretation. The theoretical analysis by Moreau and Bruna (2017) rigorously explains the source of LISTA's acceleration through matrix factorization, establishing a deep mathematical bridge between sparse coding and the implicit regularization phenomena reviewed in Section 4. Connections to advanced architectures have been formalized through ML-CSC extensions to residual networks (Zhang and Zhang 2021), EM-inspired autoencoders (Tolooshams et al. 2021), and adversarial stability bounds (Romano et al. 2020).

Our comprehensive review highlights that unrolling has matured into a universal design strategy, successfully applied across compressive sensing, video reconstruction, graph processing, MRI reconstruction (Janjušević et al. 2025), and a substantially expanded repertoire of mechanical fault diagnosis methods (Zhao et al. 2022; Zhang et al. 2022; Qin et al. 2024; Yuan et al. 2025). These architectures consistently achieve performance competitive with black-box models and offer unique transparency advantages. Practitioners can visualize learned kernels, verify feature validity, and understand failure modes. Theoretical work established rigorous connections between CNNs and hierarchical sparse coding, proving that modern architectures implicitly implement classical sparse coding principles.

Crucially, the field is expanding beyond architecture design. Recent innovations apply sparse coding principles both as a blueprint for building interpretable networks and as a diagnostic tool for analyzing existing models. Sparse autoencoders extract interpretable features from LLMs and vision Transformers, K-Deep Simplex (Tasissa et al. 2023) bridges sparse representations with manifold topology, and representational drift analysis (Ratzon et al. 2024) connects implicit sparsification to biological neural dynamics. Looking forward, integrating sparsity constraints into foundation models represents the most promising frontier for achieving high-performance AI that remains mathematically grounded and human-interpretable.

In terms of applicability, sparse representation methods are currently best suited to building interpretable-by-design architectures for signal and image processing tasks such as compressive sensing, image restoration, and fault diagnosis. They are increasingly applied to probing the internal features of large language and vision models through sparse autoencoders, rather than to settings where attention mechanisms must be modeled directly.

4 Matrix factorization methods

Matrix factorization represents one of the most fundamental mathematical frameworks connecting classical signal processing with modern deep learning. The factorization of a matrix into lower-rank components lies at the heart of dimensionality reduction, collaborative filtering, and latent factor models. In deep learning, matrix factorization emerges naturally through the analysis of linear neural networks, where the product of weight matrices corresponds to a low-rank matrix completion problem. This section reviews recent advances in understanding how deep learning implicitly incorporates matrix factorization principles, with particular emphasis on implicit regularization phenomena that drive gradient-based optimization toward low-rank solutions.

4.1 Theoretical foundations

The connection between neural networks and matrix factorization was established through the study of linear neural networks, which are neural networks without nonlinear activation functions. A depth-L linear network computing $\mathbf{y} = \mathbf{W}_L\mathbf{W}_{L-1}\cdots\mathbf{W}_1\mathbf{x}$ is mathematically equivalent to a single linear transformation $\mathbf{y} = \mathbf{W}\mathbf{x}$ where $\mathbf{W} = \prod_{l=1}^{L}\mathbf{W}_l$. Despite this equivalence in expressivity, the optimization dynamics of deep linear networks differ fundamentally from their shallow counterparts, providing a tractable setting for understanding implicit regularization in deep learning (Fig. 6a).

Bourlard and Kabil (2022) revisited the classical result that single hidden-layer autoencoders implement singular value decomposition (SVD), equivalent to principal component analysis (PCA). Their analysis demonstrates that these architectures can extract principal components corresponding to directions of maximum variance, even when nonlinearities are applied to internal representations. This foundational connection establishes that neural network training can implicitly perform matrix decomposition, providing a theoretical foundation for understanding how autoencoders organize information.

The seminal work by Razin and Cohen (2020) resolved a fundamental open question regarding implicit regularization in matrix factorization. Prior conjectures suggested that gradient descent on overparameterized matrix factorization problems implicitly minimizes nuclear norm (sum of singular values). However, Razin and Cohen refuted this conjecture, demonstrating that natural matrix factorization problems exist where implicit regularization causes unbounded norm growth. They established that *rank reduction*, rather than norm control, better characterizes the implicit bias (Fig. 6b). This insight has profound implications for understanding generalization in deep learning, with the rank-based inter-

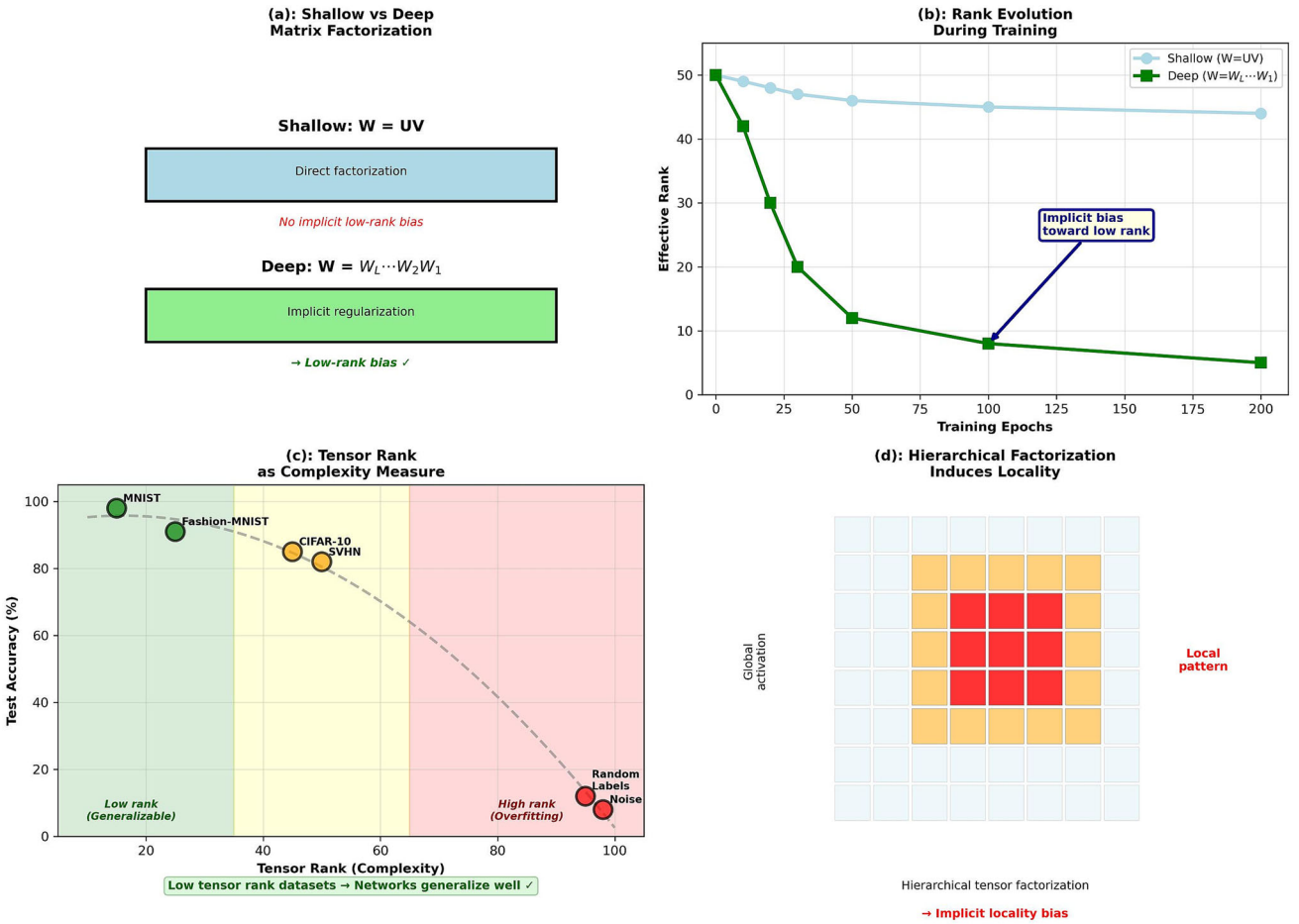


Fig. 6 Implicit regularization in matrix factorization. (**a**) Shallow vs deep factorization. Direct shallow factorization ($\mathbf{W} = \mathbf{UV}$) lacks implicit bias, while deep factorization ($\mathbf{W} = \mathbf{W}_3\mathbf{W}_2\mathbf{W}_1$) induces implicit regularization toward low-rank solutions. (**b**) Rank evolution during training. Deep networks exhibit strong implicit bias toward rank minimization during gradient descent training, whereas shallow networks show minimal rank reduction (illustrative trajectories based on Razin and Cohen 2020). (**c**) Tensor rank as complexity measure. Empirical validation that tensor rank captures dataset complexity, datasets with low tensor rank (MNIST, Fashion-MNIST) enable excellent generalization, whereas high tensor rank datasets (random labels, noise) lead to overfitting (Razin et al. 2021). (**d**) Locality bias from hierarchical factorization. Hierarchical tensor factorization induces implicit locality bias, with network activations concentrated in local spatial regions rather than distributed globally (Razin et al. 2022). These four phenomena demonstrate how architectural depth shapes implicit regularization properties and why deep learning succeeds on real-world data

pretation extending empirically to non-linear networks and potentially explaining why deep architectures generalize well.

Deep linear networks serve as a tractable model for understanding the complex dynamics of deep learning. Lampinen and Ganguli (2019) derived an analytic theory showing that gradient descent in these networks creates a singular mode detection wave, where input-output modes are acquired in order of decreasing singular value magnitude. This staged learning process explains why structured data is learned faster than randomized data. Furthermore, they demonstrated that effective transfer learning between tasks depends on the alignment of their input feature subspaces and their respective signal-to-noise ratios.

The optimization landscape of these networks was characterized by Achour et al. (2024), who identified global minimizers and differentiated between strict and non-strict saddle points. Their analysis shows that non-strict saddle points are associated with specific plateau values of the empirical risk, which can induce long training delays for first-order algorithms like ADAM. To bridge discrete and continuous perspectives, Sander et al. (2022) proved that the hidden state trajectories of ResNets approximate Neural ODE solutions, with the approximation tightening when residual functions remain smooth with depth, a property shown to be preserved by gradient descent in the linear case.

Further insights into depth were provided by Zavatone-Veth and Pehlevan (2021), who interpreted finite deep linear

Bayesian neural networks as data-dependent scale mixtures of Gaussian process predictors. They showed that increased depth induces a scale-averaging effect, allowing for a broader representation of length scales in the posterior, which improves generalization. Bernacchia et al. (2018) derived a simple, closed-form expression for the exact natural gradient in deep linear networks, proving that the natural gradient compensates for the pathological curvature of deep architectures. This led to an exponential decrease in loss and outperformed standard SGD on practical tasks like MNIST autoencoding.

4.2 Implicit regularization in deep matrix and tensor factorization

Building on the foundation of matrix factorization analysis, researchers have extended implicit regularization theory to tensor factorization and hierarchical models that more closely approximate practical deep learning architectures.

Razin et al. (2021) pioneered the theoretical examination of implicit regularization in tensor factorization, focusing on tensor completion within specific nonlinear neural network architectures. By applying a dynamical systems perspective, they successfully described the gradient descent evolution, revealing a distinct bias toward low tensor rank. This approach bypasses the traditional NP-hard complexity of tensor problems by focusing on the optimization trajectory rather than the static landscape. Motivated by this, they validated tensor rank as an effective complexity measure, finding it strongly correlates with the datasets where neural networks exhibit good generalization (Fig. 6c).

Extending this line of work, Razin et al. (2022) investigated implicit regularization in *hierarchical* tensor factorization, which mathematically mirrors the structure of deep convolutional neural networks. By leveraging dynamical systems theory, they surmounted the obstacles of hierarchical structures to demonstrate an implicit regularization toward low hierarchical rank. Crucially, they showed this manifests as a bias toward *locality* in CNNs (Fig. 6d). While global structures are theoretically possible, the optimization naturally favors local feature combinations. The authors developed an explicit regularizer to counteract this locality bias. This improved CNN performance on non-local tasks without requiring architectural changes, challenging conventional assumptions.

Recent work has further characterized the dynamics of implicit bias toward low-rank under practical training conditions. Chou et al. (2024) analyzed gradient descent dynamics in deep matrix factorization, establishing that the effective rank of iterates monotonically increases after an initial phase, and that early stopping extracts low-rank regularized solutions useful for denoising. Chatterji and Long (2023) proved that deep linear networks exhibit benign overfitting by matching the conditional variance of the minimum ℓ_2-norm interpolant, showing that depth neither improves nor degrades the ability to hide noise relative to shallow networks. Tovey et al. (2023) demonstrated that the starting entropy and trace of the empirical NTK are predictive of final generalization performance, using these collective variables to develop an optimal data selection strategy via random network distillation. Ledent and Alves (2024) established generalization bounds for deep non-linear matrix completion with Schatten p quasi-norm constraints, extending implicit regularization guarantees from linear to nonlinear factorization regimes.

Soltanolkotabi et al. (2025) explored overparameterized asymmetric matrix sensing, moving beyond the symmetric positive semi-definite (PSD) constraints of previous studies. They proved that factorized gradient descent from small random initialization recovers the underlying low-rank matrices from linear measurements whose sensing operator satisfies the Restricted Isometry Property. Their analysis identified two key mechanisms that is implicit balancing, where factor trajectories are coupled to maintain similar magnitudes throughout training, and an inherent algorithmic regularization that favors low-rank models without any explicit nuclear norm penalty. Kumar and Haupt (2025) studied gradient flow dynamics of homogeneous networks beyond the early training phase, proving that after weights escape the origin their directional convergence is maintained. They showed that the sparsity structure formed during the initial phase is preserved throughout the subsequent trajectory until the network reaches its first saddle point, extending implicit regularization guarantees beyond the infinitesimal regime near initialization.

Du et al. (2018) proved that gradient flow in deep homogeneous models automatically balances layer norms without explicit regularization. Specifically, differences between squared Frobenius norms across layers remain invariant throughout training. Chen and Chen (2020) proved that overparameterizing the matrix factorization model is equivalent to applying SGD with momentum and an adaptive learning rate on the standard model. This provides a rigorous explanation for why overparameterization accelerates convergence without changing model expressiveness. Blanc et al. (2020) characterized implicit regularization near zero-training-error solutions for networks trained with label noise. They proved that the implicit regularizer penalizes the sum of squared ℓ_2 gradient norms across data points. This effect holds universally across architectures, widths, and depths, driving networks toward simpler solutions such as minimum-rank matrices in matrix sensing.

Recent work (Ghosh et al. 2025) analyzed deep matrix factorization beyond the edge of stability (EOS) regime, where the largest Hessian eigenvalue equilibrates about the stability threshold $2/\eta$. For deep linear networks, loss oscil-

lations beyond EOS follow a period-doubling route to chaos restricted to a low-dimensional subspace determined by the learning rate. The symmetry-induced conservation laws that normally balance weight magnitudes across layers collapse at EOS, driving the transition into chaotic dynamics. The EOS regime itself was analyzed by Arora et al. (2022), who proved that in deterministic gradient descent, sharpness stabilizes near $2/\eta$ through an implicit regularization that minimizes the top eigenvalue of the Hessian. Achour et al. (2024) identified that non-strict saddle points in the loss landscape create gradient flow plateaus, providing a second-order perspective on how singular value alignment directs the network toward global optimality despite the non-convex landscape.

The role of stochastic optimization in implicit regularization has been characterized by several complementary analyses. Damian et al. (2021) proved that SGD with label noise converges to a stationary point of a regularized loss $\tilde{L}(\theta) = L(\theta) + \lambda R(\theta)$, where λ depends inversely on batch size and $R(\theta)$ explicitly penalizes sharp minimizers through the eigenvalues of the Hessian. Smaller batches and larger learning rates strengthen this regularization, pushing the network toward flatter solutions that generalize better.

Cavazza et al. (2018) proved that dropout in matrix factorization is equivalent to a deterministic model regularized by products of squared Euclidean norms. For variable-size factorizations, dropout finds global minima of convex approximation problems with squared nuclear norm regularization, effectively acting as a low-rank regularizer with data-dependent singular-value thresholding. Mianjy and Arora (2019) extended this analysis by deriving a closed-form implicit regularizer for dropout in matrix factorization. They showed that dropout jointly penalizes the Frobenius norms of both factor matrices and encourages balanced singular value distributions, a property that promotes both low-rank and generalization. Ergen et al. (2022) analyzed batch normalization through convex duality. They obtained exact convex representations of weight-decay regularized ReLU networks with batch normalization (BN), trainable in polynomial time. Their analysis also shows that gradient descent induces an algorithmic bias effect on non-convex BN networks, with the implicit regularization aligned with low-rank biases predicted by matrix factorization theory.

Sahiner et al. (2021) established a significant link between neural network training and copositive programming. They showed that the non-convex training of two-layer vector-output ReLU networks is equivalent to a finite-dimensional convex copositive program. This implies that neural networks implicitly solve copositive programs through *semi-nonnegative matrix factorization*. Based on this, the authors derived polynomial-time algorithms for finding global minima under fixed data rank conditions, and in specific cases, showed that the global optimum is recoverable via soft-thresholded SVD.

Neural collapse has been analyzed through the matrix factorization lens. Dang et al. (2023) proved that for deep linear networks, neural collapse emerges in global minimizers, with class imbalance producing a modified collapse geometry of orthogonal vectors that deviates predictably from the symmetric simplex structure. This aligns the terminal phase of training with the low-rank inductive biases found in matrix factorization theory, providing a unified view of how networks simplify their representations at convergence.

Bridging model-based optimization and deep learning, Ma et al. (2019) proposed the deep tensor ADMM-Net for snapshot compressive imaging. Instead of using black-box approximations, they unfolded the iterative tensor ADMM algorithm into a deep network structure defined by explicit tensor operations. A critical innovation is that the network learns the low-rank tensor representation domain via end-to-end training rather than relying on a pre-defined sparse basis. This approach yields a mathematically interpretable model that achieves 0.8-2.5 dB PSNR improvement and massive speedups (1500×-3600×) over traditional methods.

4.3 Non-negative matrix factorization for interpretability

Non-negative matrix factorization (NMF) imposes non-negativity constraints on factor matrices, yielding parts-based representations particularly suited for interpretability. Deep NMF architectures extend this principle hierarchically by stacking NMF layers. Networks thereby obtain hierarchical representations that combine interpretability with increased expressiveness (Chen et al. 2022). Guo and Zhang (2020) extended this framework with explicit sparsity penalties within each NMF layer. They showed that sparse deep NMF produces sparser, more disentangled basis vectors at each hierarchical level, with improved parts-based interpretability for face analysis tasks. Recent work has leveraged NMF to explain DNN decisions through concept-based decomposition.

Zhang et al. (2021a) proposed the invertible concept-based explanation (ICE) framework using non-negative concept activation vectors (NCAVs). The framework addresses CNN interpretability by decomposing mid-layer feature maps using matrix factorization methods. When evaluated against metrics of fidelity (approximation to target models) and interpretability (meaningfulness to humans), NCAVs derived via NMF provided superior performance compared to other factorization approaches in both computational and human-subject experiments. This framework enables extracting meaningful concept-level explanations for pre-trained CNNs at both instance-level and class-level.

In the audio domain, Parekh et al. (2024) and Akman and Schuller (2024) utilized NMF to generate listenable explanations. Parekh et al. addressed audio classification by training

an interpreter to generate intermediate embeddings as time-activations of a pre-learned NMF dictionary. This produces audible reconstructions of the features most relevant to the network's decision. Akman et al. extended this to transformer models with AttHear, combining attention mechanisms with NMF to compute time activation for spectral patterns. This addresses the gap in explainability methods that predominantly focus on visual and textual explanations, providing crucial audio-modality interpretations.

NMF-based interpretability extends naturally to clustering and community detection tasks. Gai and Liu (2023) proposed a sparse orthogonal NMF framework integrated into an interpretable neural network (NINN) for multimedia clustering. By treating the clustering layer as a differentiable reconstruction problem, the architecture ensures direct interpretability rather than relying on post hoc explanations. The orthogonality constraint facilitates non-overlapping cluster assignments, while sparsity promotes localized features, allowing practitioners to inspect specific basis atoms driving cluster decisions. He et al. (2022) extended this to community detection in complex networks, combining graph attention autoencoders with NMF to capture nonlinear structures while preserving the parts-based interpretability of factorized representations. Furthermore, Huang et al. (2022) proposed Deep AE-like NMF for multi-view data representation, stacking NMF layers in an AE architecture that jointly learns consistent and view-specific representations across modalities while maintaining NMF's interpretability guarantees.

Extending NMF-based interpretability to physical mixture modeling, Xiong et al. (2022a) developed SNMF-Net for hyperspectral unmixing. The network unfolds the L_p sparsity constrained NMF (L_p-NMF) optimization algorithm into an end-to-end architecture with alternating structure. By seamlessly linking network parameters and submodules with the L_p-NMF alternating optimization steps, SNMF-Net achieves high physical interpretability while integrating prior knowledge on unmixing and sparse representation theory. This demonstrates how NMF principles can guide network design for domain-specific problems requiring interpretable constituent decomposition.

4.4 Applications

Matrix factorization principles have been integrated into deep learning architectures for specific application domains, particularly where interpretability and physical constraints are important.

4.4.1 Algorithm unfolding for hyperspectral and remote-sensing imaging

Algorithm unfolding, mapping iterative optimization algorithms to neural network layers, has emerged as a powerful paradigm for building interpretable-by-design networks, particularly in HSI where physical models provide strong inductive biases.

Several architectures embed low-rank constraints directly into network structure. The recurring design pattern is to unfold a decomposition model into network layers and learn its parameters end-to-end. The methods differ in the decomposition they impose, namely low-rank representation, low-rank plus sparse, subspace-multidimensional sparsity, non-negative factorization, or coupled factorization across sensors. Each choice reflects the physical structure of the target problem.

Li et al. (2023) proposed LRR-Net, which maps the iterative ADMM solution of the low-rank representation model into neural network layers for hyperspectral anomaly detection, removing manual parameter selection while retaining effective background-target separation. In a related application, Li et al. (2026) proposed LSNet, a low-rank- and sparse-aware deep unfolding network for infrared small target detection. It unfolds a low-rank-plus-sparse decomposition so that the background is recovered as a low-rank component and small targets as a sparse component, extending the interpretable-by-design unfolding paradigm to detection tasks in heterogeneous backgrounds. Similarly, Xiong et al. (2022b) developed SMDS-Net by unfolding a subspace-based multidimensional sparse model to exploit the inherent spatial redundancies and spectral low-rank properties of hyperspectral data. Earlier foundational work by Fang et al. (2018) established sparsity-constrained deep NMF for hyperspectral unmixing, demonstrating that combining non-negativity with sparsity constraints within a deep architecture yields physically interpretable endmember extraction with improved spectral purity.

For multi-sensor fusion, Wu et al. (2024) proposed CMF-FUSnet, mapping iterative CMF optimization into a dual-branch network where one branch extracts endmembers from hyperspectral data and the other estimates abundances from multispectral data, with reconstruction generated by multiplying learned components. Extending matrix factorization to physics-constrained settings, Tetali et al. (2024) proposed wave physics-informed matrix factorizations that embed wave propagation constraints directly into factor matrices, ensuring physically consistent decompositions for acoustic and seismic data while maintaining the interpretability of classical NMF. Addressing high-dimensional visual data, Xu et al. (2025) developed a parameterized low-rank regularizer that uses neural networks to parameterize the factor matrices, imposing an automatically learned penalty that enhances generalization across diverse inverse problems including HSI denoising and inpainting.

The consistent lesson across these unfolded factorization models is that embedding the physically appropriate decomposition into the architecture can deliver interpretability with

little or no performance penalty. The open difficulty is not interpretability but choosing or learning the correct rank and sparsity structure for a given sensing problem, which remains largely problem-specific.

4.4.2 Universal feature learning

Huang et al. (2024) developed a comprehensive information-theoretic framework connecting classical dimensionality reduction methods (SVD, canonical correlation analysis, PCA) to modern learning systems including neural network architectures. Their analysis reveals fundamental theoretical links among these methods, showing how they provide universal principles for extracting low-dimensional features from high-dimensional data. This unifying perspective offers insights into softmax regression, matrix factorization approaches, and semi-supervised learning architectures. This information-theoretic view also connects to the implicit low-rank and manifold structure discussed in this review. The information bottleneck principle (Tishby and Zaslavsky 2015) characterizes learning as a progressive compression of input representations, and the resulting reduction in effective dimensionality parallels the rank minimization and low-dimensional manifold structure that matrix factorization and manifold learning identify through different mathematical frameworks. Applications of these classical factorization methods span diverse domains including biomedical signal analysis (Khalid and Seghouane 2014b; Shah et al. 2012; Khalid et al. 2012), and computer vision (Ma et al. 2019; Zhang et al. 2021a).

Beyond signal processing, matrix factorization has proven effective for geometric computer vision problems. Tejus et al. (2023) formulated rotation synchronization as an unsupervised deep matrix factorization problem over the special orthogonal group SO(3). The objective is to recover consistent absolute rotations from noisy pairwise measurements. This approach demonstrates that matrix factorization interpretability and implicit low-rank regularization extend naturally to non-Euclidean settings in structure-from-motion and Simultaneous Localization and Mapping (SLAM) pipelines. In the collaborative filtering domain, Xu et al. (2021a) provided a theoretical reconciliation between neural collaborative filtering (NCF) and classical matrix factorization (MCF). They proved that overparameterized NCF and MCF have similar limiting expressivity as kernelized predictors. Their analysis further reveals that gradient descent induces a max-margin inductive bias, with MCF more effective for transductive tasks and NCF excelling in inductive settings.

Deng and Xiao (2023) applied low-rank tensor completion to hyperparameter optimization via meta-learning. They demonstrated that representing the hyperparameter performance space as a tensor preserves essential spatial correlations, allowing the underlying low-rank structure to enable efficient automatic recommendations across large-scale tasks. This work effectively extends implicit low-rank regularization principles to the domain of automated neural architecture search and model configuration.

4.5 Critical analysis

The body of work on implicit regularization in matrix factorization has produced significant theoretical insights, yet several limitations remain.

Gap between theory and practice Most analyses focus on linear networks or simplified nonlinear models. While providing mathematical tractability, the extent to which insights transfer to practical architectures with complex nonlinearities, normalization, and residual connections remains unclear. The hierarchical tensor factorization results (Razin et al. 2022) and neural collapse analysis (Dang et al. 2023) represent progress, but significant work remains.

Optimization assumptions Implicit regularization results typically require specific initialization schemes (near-zero or small random) and learning rates. Practical training employs different strategies (Glorot and Bengio 2010; He et al. 2015), adaptive learning rates, stochastic gradients, and momentum. The SGD implicit regularization results (Damian et al. 2021) and gradient flow dynamics analysis (Kumar and Haupt 2025) have begun to address stochasticity and post-initialization effects, but a complete theory remains elusive.

Explicit regularization interactions The relationship between implicit and explicit regularization remains underexplored. Although dropout connects to nuclear norm regularization (Cavazza et al. 2018; Mianjy and Arora 2019) and batch normalization to structured matrix decompositions (Ergen et al. 2022), how weight decay and layer normalization interact with implicit low-rank biases requires further investigation.

Interpretability limitations NMF-based methods (Zhang et al. 2021a; Parekh et al. 2024; Akman and Schuller 2024) show promise but face scalability challenges for large models. The faithfulness of concept-based explanations to actual model computations requires continued scrutiny.

4.6 Summary

This section reviewed 46 papers connecting matrix factorization to DNN interpretability. The theoretical foundations establish that deep linear networks and linear autoencoders exhibit SVD/PCA-like structure, while gradient descent in matrix factorization exhibits an implicit bias toward low-rank

solutions that is not fully captured by standard norm minimization. Analytic theories of learning dynamics (Lampinen and Ganguli 2019) reveal sequential feature acquisition governed by data covariance, explaining both generalization and transfer learning. Extensions to tensor factorization and hierarchical models reveal how implicit regularization induces locality bias in convolutional networks (Razin et al. 2022). The role of step size and stochasticity in shaping the implicit bias has been clarified through SGD theory (Damian et al. 2021) and gradient flow dynamics beyond initialization (Kumar and Haupt 2025). Recent results on benign overfitting (Chatterji and Long 2023), edge-of-stability dynamics (Ghosh et al. 2025; Arora et al. 2022), and neural collapse (Dang et al. 2023) clarify how rank minimization bias operates across the full training trajectory and into the terminal phase. Dropout and batch normalization have been connected to nuclear norm and structured matrix decomposition regularization (Cavazza et al. 2018; Mianjy and Arora 2019; Ergen et al. 2022). NMF enables concept-based explanations in vision, audio, and clustering domains, while applications in HSI, physics-informed factorization, geometric computer vision, and collaborative filtering demonstrate the breadth of matrix factorization principles in deep learning. The key insight from this literature is that gradient-based optimization on factorized models provides an inherent bias toward low-rank, interpretable solutions that can be theoretically characterized and practically exploited across a wide range of architectures and domains.

In terms of applicability, matrix factorization methods are currently best suited to explaining implicit bias and low-rank structure in linear and tensor models, and to concept-based interpretability through NMF in vision, audio, and clustering tasks. They apply less directly to fully nonlinear architectures with normalization and residual connections, where the theory remains incomplete.

5 Manifold learning methods

Manifold learning constitutes a smaller body of interpretability-focused work than the other three frameworks. We nonetheless treat it as a distinct fourth pillar. It provides geometric tools, including curvature, intrinsic dimension, and optimal-transport structure, that the kernel, sparse, and factorization perspectives cannot express. Its inclusion therefore reflects theoretical distinctness rather than corpus size.

Manifold learning provides a geometric framework for understanding deep neural networks through low-dimensional structure in high-dimensional data. The manifold hypothesis that real-world data lies on or near low-dimensional manifolds offers tools for analyzing neural representations, understanding generalization, and explaining phenomena like adversarial vulnerability. Recent work has expanded this framework in two important directions i) optimal transport (OT) theory provides rigorous tools for understanding how neural networks transform probability distributions supported on manifolds, and ii) approximation theory establishes precise bounds showing that network expressiveness scales with manifold dimension rather than ambient dimension.

5.1 Theoretical foundations

Cohen et al. (2020) established foundational connections between manifold geometry and classification capacity in deep neural networks. They analyzed how object manifolds transform across network hierarchies. As data progresses through the network, object manifolds undergo three coordinated changes. These are compression in radius, reduction in dimensionality, and decreased correlations between classes. These geometric evolutions drive improvements in classification capacity (Fig. 7). Their theoretical framework quantifies how these three geometric properties determine linear separability, revealing that deep networks progressively untangle object manifolds through coordinated geometric transformations. This work provides rigorous mathematical tools for understanding representation learning as geometric manifold reformatting.

The manifold hypothesis states that although data may exist in high dimensional spaces, the underlying structure occupies far fewer dimensions. Kiani et al. (2024) investigated the computational complexity of learning under this hypothesis. They proved that learning remains fundamentally difficult even when data lies on smooth manifolds, extending hardness results from statistical query and cryptographic settings. However, they also showed that specific geometric conditions render the problem learnable, particularly when the manifold volume is constrained or when the manifold is reconstructible via classical algorithms. This indicates that manifold structure alone is insufficient for efficient learning; additional constraints on volume or reconstructibility are necessary.

Complementing these computational complexity results, recent approximation theory provides rigorous bounds on how well deep networks can represent functions defined on smooth manifolds. Labate and Shi (2024) proved that for Hölder functions defined on a d-dimensional smooth manifold $\mathcal{M}$ embedded in $\mathbb{R}^D$ with $d \ll D$, deep ReLU networks achieve uniform convergence of both approximation and generalization errors. Crucially, these error bounds depend only on the intrinsic manifold dimension d rather than the ambient dimension D. Specifically, the multiplicative constant in these bounds is independent of D, providing a theoretical foundation for understanding why deep networks escape the curse of dimensionality.

Building an explicit architectural realization of this manifold-adaptive learning, Chui et al. (2018) proposed a deep

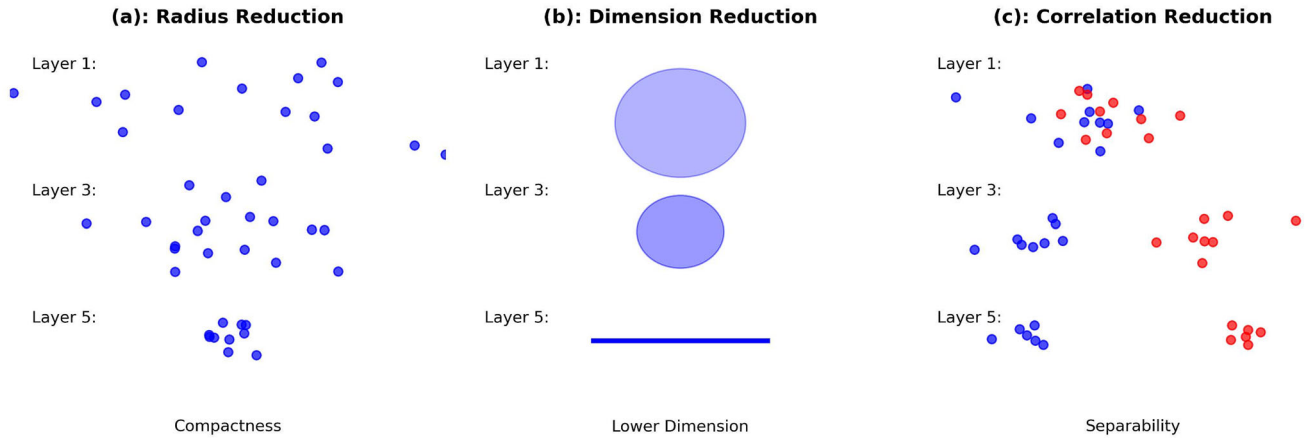


Fig. 7 Geometric transformation of object manifolds across deep network layers (Cohen et al. 2020). (**a**) Radius reduction: manifolds become progressively more compact across layers. (**b**) Dimensionality reduction: intrinsic manifold dimension decreases from high dimensional (ellipse) to low dimensional (line). (**c**) Correlation reduction: inter manifold correlations decrease as class manifolds (blue vs red) separate, enabling linear classification. These three coordinated geometric changes explain how deep networks achieve high classification capacity through progressive manifold untangling

network framework for localized manifold learning using Heaviside and square-rectifier activations, in which each hidden layer is assigned a specific and interpretable learning task. In their three-layer model, the first layer performs dimensionality reduction, the second layer reduces bias, and the third layer reduces variance, with a feedback component for outlier handling. Their central theoretical result establishes an approximation rate of $O(m^{-2s/(2s+d)})$ for regression functions of regularity s, where the unknown manifold dimension d replaces the ambient dimension D of standard shallow network bounds. This architecture makes the manifold learning process transparent and modular, with each layer's contribution to generalization mathematically characterized. This echoes the layer-by-layer geometric transformation picture established by Cohen et al. (2020).

Chui and Mhaskar (2018) developed a complementary approach, constructing deep networks that find local coordinate systems for unknown manifolds without eigendecomposition. Their method reduces function approximation in a high-dimensional ambient space to classical approximation on a low-dimensional cube, with a priori error bounds that do not require prior knowledge of the target function's smoothness. Unlike backpropagation-based methods, their construction provides universal approximation guarantees that adjust locally and automatically to the smoothness of the target function.

Psenka et al. (2024) proposed flattening networks (FlatNet) for representation learning based on manifold flattening. Their approach builds neural networks layer by layer, where each layer progressively flattens curved data manifolds through a smooth geometric transformation process, reconstructing the manifold structure from available data samples. Unlike standard autoencoders which do not inherently exploit data's geometric structure, FlatNet explicitly maps curved data distributions into lower-dimensional linear spaces while preserving essential topological features. This approach achieves a balance of mathematical transparency and scalability often missing in traditional manifold learning methods.

Complementing these theoretical foundations, Buchanan et al. (2021) analyzed the multiple manifold problem, binary classification of low dimensional submanifolds, through the NTK lens. They proved that when network depth is large relative to geometric and statistical properties of the data, and width grows polynomially with depth, gradient descent rapidly learns to separate manifolds with high probability. Their analysis demonstrates concrete benefits of depth (as a fitting resource enabling smoother separating functions) and width (as a statistical resource enabling concentration). The work provides essentially optimal concentration rates for the NTK of deep networks over submanifolds, contributing rigorous theoretical understanding of how architectural choices affect manifold separation capacity.

These geometric perspectives do not stand apart from the other frameworks reviewed in this work. They connect to them in concrete mathematical ways. The analysis of Buchanan et al. already illustrates one such link, since it studies manifold separation through the NTK. More broadly, manifold learning and NTK theory both examine the geometry that networks learn, with NTK working through the kernel spectrum and manifold learning through representational structure. Manifold learning also shares a foundation with matrix factorization. The implicit bias toward low-rank solutions studied in Section 4 corresponds closely to the principle that data concentrates on a low-dimensional manifold, since the rank of a learned representation reflects its intrin-

sic dimension. We develop these connections in detail in Section 6. We note them here so that the geometric tools introduced in this section are understood as part of a connected theoretical picture rather than as an isolated approach.

5.2 Optimal transport and manifold-based geometric frameworks

Recent work has formalized the connection between manifold learning and OT theory, providing new geometric tools for understanding deep generative models and scientific applications.

Lei et al. (2020) introduced the manifold distribution principle as a formal geometric foundation for deep learning. They showed that natural data concentrates near low-dimensional manifolds, and that deep generative models accomplish two distinct tasks, namely manifold learning and probability distribution transformation. Analyzing GANs through this OT lens, they proved that the generator computes the OT map while the discriminator computes the Wasserstein distance, and that both reduce to convex geometric optimization. This OT perspective reveals the generator and discriminator as fundamentally collaborative rather than adversarial, and explains mode collapse as a failure of the OT map to cover all components of the data manifold. Based on this analysis, they proposed the AE-OT model, in which an AE performs manifold learning and an explicit OT map handles distribution transformation, improving theoretical transparency and computational stability.

Lu et al. (2023) extended manifold learning from representation analysis to scientific discovery by reformulating conservation law identification as a manifold learning problem. Conservation laws constrain trajectories to low-dimensional level-set manifolds in state space, so identifying conserved quantities is equivalent to discovering the intrinsic geometry of trajectory data. Their non-parametric approach uses OT theory to extract conserved quantities directly from data, without requiring an explicit system model or accurate time information. The identified conservation laws correspond to geometrically meaningful invariants of the dynamics, making the method interpretable by construction. This application demonstrates the dual utility of manifold learning as both a theoretical lens for understanding neural networks and a practical tool for interpretable scientific modeling.

5.3 Manifold geometry of neural representations

The studies in this subsection share a common strategy. Each computes a geometric descriptor of the learned representation, such as curvature, intrinsic dimension, or topological complexity, and relates it to a behavioral property of the network, such as class bias, forecasting accuracy, or generalization. They differ in which descriptor they track and which behavior it predicts, and read most usefully as a collective demonstration that representational geometry is a measurable correlate of model behavior rather than as isolated case studies.

Ma et al. (2023) analyzed manifold curvature in neural representations for long-tailed classification. They discovered that during training, class-wise accuracy becomes increasingly correlated with curvature properties rather than simple manifold separability, revealing that unbalanced curvature distribution, not merely class imbalance, drives model bias. Networks develop feature manifolds with heterogeneous curvature: high curvature characterizes difficult classes while easy classes exhibit flatter geometry. Their proposed curvature regularization promotes more uniform curvature distributions across classes, improving performance on both long-tailed and sample-balanced datasets.

Examining manifold learning as a deep learning method itself, Pai et al. (2019) proposed deep isometric manifold learning (DIMAL), a fully unsupervised approach for computing distance-preserving maps that generate low-dimensional embeddings. Using a Siamese network configuration to solve least squares multidimensional scaling, they train neural networks to approximately preserve geodesic distances. Training with only sparse landmark samples, DIMAL achieves significantly improved local and nonlocal generalization compared to classical nonparametric manifold learning methods. Despite being primarily methodological, this work provides a framework for geometric analysis of network architectures by integrating deep learning with multidimensional scaling. It offers preliminary insights into how networks learn generalizable representations.

Kaufman and Azencot (2024) applied manifold learning to analyze deep transformer models for time series forecasting. Assuming latent representations lie near low dimensional manifolds, they tracked geometric metrics such as intrinsic dimension and principal curvature. Their analysis indicates that transformers maintain consistent geometric structures throughout their depth, and these structural properties correlate strongly with final model performance. Untrained models initially exhibit disparate manifold structures, but these rapidly converge within a few epochs. Geometric properties measured after this early convergence phase correlate with final accuracy, enabling the identification of promising architectures without the computational cost of full training.

Magai et al. (2025) investigated foundation model representations from a geometric perspective. To understand how knowledge is encoded in embedding spaces, they analyzed the evolution of embeddings across network layers by applying log-Euclidean signature distances to measure point cloud similarities. They introduced the domain geometry divergence approach for quantifying differences in visual image domains, conducting experiments with foundation models

including CLIP, DINOv2, BeiT, and MAE. Their analysis reveals that the geometric structure of the learned manifold is heavily influenced by the specific pretraining objectives, with self-supervised and supervised models developing distinct topological characteristics.

Magai (2023) investigated DNN representations from a topological and geometric perspective. To understand how knowledge is encoded, they analyzed the evolution of data manifolds across network layers using topological data analysis (TDA) and persistent homological fractal dimension (PHdim). Their analysis reveals significant differences in how architecture families transform data. CNNs exhibit a hump-shaped evolution of intrinsic dimensionality due to their pyramidal hierarchical structure, whereas Vision Transformers maintain a more consistent topological structure across all depth levels. Furthermore, Magai demonstrated that the gradual simplification of these topological descriptors during training serves as a reliable predictor of a model's generalization ability.

Zhang et al. (2024b) developed a theory of unimodal bias in multimodal deep linear networks, analyzing how architecture and data statistics shape the geometry of learned representations across modalities. They calculated the duration of the unimodal phase as a function of fusion depth, dataset statistics, and initialization, showing that deeper fusion leads to longer unimodal phases. A long unimodal phase can produce permanent generalization deficits in overparameterized networks. Their analysis connects fusion architecture choices directly to the geometric properties of learned representations, extending deep linear network theory to the multimodal setting.

Across these studies a consistent picture emerges. Geometric descriptors measured early in training, particularly intrinsic dimension and curvature, are predictive of final accuracy and bias, which makes manifold geometry not only a descriptive lens but a diagnostic one. The shared limitation is that these descriptors are estimated from finite samples without uncertainty quantification, so their predictive value, while empirically consistent, currently rests on weaker theoretical guarantees than the other three frameworks.

5.4 Applications

Zhang et al. (2020a) investigated adversarial examples through manifold geometry. They introduced the adversarial region concept, characterizing adversarial perturbations as vectors orthogonal to the data manifold's tangent space. This geometric interpretation provides a principled explanation for why adversarial examples transfer across models. Building on this insight, they developed a target-free PCA-based generation method that eliminates the need for pre-trained classifiers, requiring only the data distribution itself while achieving competitive transferability. This reveals that adversarial vulnerability stems from data geometry rather than model-specific artifacts.

The OT-based manifold framework of Lei et al. (2020) provides interpretable understanding of generative model failure modes beyond adversarial settings. By characterizing mode collapse as a geometric breakdown in the OT map, the framework predicts when and why generative models will fail to cover the full data manifold, an interpretability property absent from standard generative adversarial network analysis. Similarly, the conservation law discovery framework of Lu et al. (2023) demonstrates that manifold learning provides actionable scientific insights: identified conserved quantities enable building stable, interpretable predictive models of physical systems that remain valid across unseen dynamical regimes.

5.5 Critical analysis

Manifold hypothesis validity The assumption that data lies on low-dimensional manifolds remains formally unverified for most real datasets. How violations affect interpretability methods requires investigation. The approximation theory results of Labate and Shi (2024) provide bounds conditional on the manifold hypothesis holding, but do not verify whether it holds in practice.

Computational intractability Hardness results (Kiani et al. 2024) establish that learning remains hard despite manifold structure, raising questions about achievable interpretability. The non-parametric OT and manifold approaches of Lu et al. (2023) scale well for physical systems with low state-space dimension but face challenges for high-dimensional neural representations.

Manifold estimation challenges Estimating manifold structure from finite samples introduces errors lacking rigorous uncertainty quantification. The sparse geodesic sampling of DIMAL (Pai et al. 2019) mitigates but does not eliminate this challenge.

Scalability Geometric analysis of large-scale models (billions of parameters) presents computational challenges; manifold structure may be too complex for current techniques. The layer-assignment architecture of Chui et al. (2018) offers modular scalability but has not been validated at the scale of modern foundation models.

Theory-practice gap Theoretical results assume idealized conditions. The approximation bounds of Labate and Shi (2024) hold for smooth manifolds and Hölder functions; transfer to practical deep learning with stochastic optimization, batch normalization, and non-smooth loss surfaces requires further validation.

5.6 Summary

This section reviewed 17 papers connecting manifold learning to DNN interpretability. Cohen et al. (2020) established the foundational connection between manifold geometry and classification capacity. Approximation theory shows that deep ReLU networks achieve generalization rates governed by intrinsic dimension d rather than ambient dimension D (Labate and Shi 2024), with the layer-wise architecture of Chui et al. (2018) making this transparent through explicit layer roles. Chui and Mhaskar (2018) extended this with backpropagation-free deep nets achieving universal approximation on unknown manifolds through local coordinate construction. NTK analysis reveals how depth and width enable manifold separation (Buchanan et al. 2021), while manifold flattening provides interpretable-by-design networks (Psenka et al. 2024). The OT perspective of Lei et al. (2020) unifies manifold learning with generative modeling, and Lu et al. (2023) extend this to scientific discovery through conservation law identification. Geometric analysis reveals curvature as a driver of long-tail bias (Ma et al. 2023), transformer performance (Kaufman and Azencot 2024), and adversarial vulnerability (Zhang et al. 2020a). Zhang et al. (2024b) show that fusion depth governs the duration of unimodal phases in multimodal deep linear networks, connecting architecture choices to representation geometry. Foundation model representations can be analyzed through geometric divergence measures (Magai et al. 2025), while DIMAL provides geometric insights into network generalizability (Pai et al. 2019). The key insight is that manifold geometry offers unique interpretability tools revealing geometric structure in neural networks not visible through other classical frameworks.

In terms of applicability, manifold learning is currently best suited to analyzing representation geometry, class separability, and curvature, to understanding generative models through optimal transport, and to scientific-discovery settings with low-dimensional state spaces, rather than to high-dimensional representations where the manifold hypothesis is difficult to verify.

6 Conclusion

This comprehensive review has systematically examined how classical ML frameworks, NTK methods, sparse representations, matrix factorization, and manifold learning, provide theory-grounded lenses for interpreting learning dynamics and representation formation. Through analysis of 203 papers published between 2016 and 2026, we have evaluated the state of classical approaches to interpretability, identified fundamental limitations, and charted directions for future progress. We conclude by synthesizing key findings, offering concrete recommendations, and reflecting on the broader role of theory in achieving trustworthy AI.

6.1 Summary of key findings

Before synthesizing key findings, Table 8 provides a concise comparison of all four classical frameworks across the most critical dimensions examined in this review.

6.1.1 The promise and limitations of classical theory

Classical ML frameworks have delivered significant theoretical insights into deep learning, yet confront fundamental limitations that constrain practical applicability.

Our review reveals four major accomplishments using classical theory. First, NTK theory transforms neural network optimization from intractable non-convex parameter space problems into tractable convex function space problems with provable convergence guarantees and generalization bounds (Jacot et al. 2018) establishing *rigorous characterization of training dynamics*. Second, algorithm unrolling demonstrates that embedding classical optimization algorithms in network structures produces interpretable yet high-performing systems across image processing (Wang et al. 2020a), compressive sensing (Li et al. 2024), and mechanical diagnosis (An et al. 2022) leading to *interpretable-by-design architectures*. Third, matrix factorization theory proves that gradient descent exhibits implicit regularization toward rank minimization (Razin and Cohen 2020), tensor factorization (Razin et al. 2021), and locality in hierarchical models (Razin et al. 2022), which reveals *fundamental bias characterization* underlying deep learning success. Fourth, manifold learning provides mathematical tools for understanding representation learning as coordinated geometric transformations that reduce manifold radius, dimensionality, and correlations (Cohen et al. 2020) providing *geometric understanding*.

Despite these achievements, fundamental gaps prevent complete theoretical understanding of practical deep learning. First, *the lazy-rich divide* where NTK theory's rigorous guarantees require large initialization or infinite width, producing lazy regime behavior where networks minimally update features (Jacot et al. 2018; Woodworth et al. 2020). However, practical networks operate in the rich regime with small initialization, enabling feature learning that transcends kernel regime predictions and breaks theoretical guarantees (Chen et al. 2020a; Allen-Zhu et al. 2019). Second, *architectural restrictions* where most theoretical results apply to idealized architectures (fully-connected networks, linear networks, simple CNNs), while modern systems incorporate components (normalization, residuals, attention) violating theoretical assumptions. Third, *computational intractability* where classical analyses require prohibitive computation

Table 8 Comparison of classical ML frameworks for deep learning interpretability across key dimensions

Dimension	NTK Methods	Sparse Representations	Matrix Factorization	Manifold Learning
Core Theory	Kernel methods in RKHS	ℓ_1 optimization	Low-rank decomposition	Differential geometry and OT
Main Strength	Rigorous convergence guarantees	Interpretable-by-design architectures	Explains implicit regularization	Geometric visualization; OT-based generative modeling; approximation bounds on manifolds
Main Weakness	Lazy-rich regime divide	Approximation to true optimization	Limited to linear/tensor models	Manifold hypothesis unverified
Gap to Practice	Requires infinite width or large initialization, whereas real networks are finite with small initialization	Networks deviate from theoretical sparse codes	Most results for linear neural networks only	Manifold estimation from finite data; approximation bounds assume smooth manifolds
Validity Conditions	Infinite width; lazy regime ($\Theta(t) \approx \Theta(0)$); positive definite NTK; squared loss; gradient flow	Training distribution; sufficient sparsity; well-conditioned dictionary	Linear or diagonal networks; small random initialization; gradient flow	Data lies on low-dimensional manifold; sufficient samples for estimation; Hölder smoothness for approximation bounds
Failure Modes	Finite-width feature-learning networks; rich regime; non-squared losses	Out-of-distribution inputs; overcomplete dictionaries; strong nonlinearity	Nonlinear architectures; normalization layers; residual connections	High-dimensional manifolds; manifold hypothesis violated; scarce data
Interpretability Type	Training dynamics; convergence guarantees; architectural bias	Feature decomposition; algorithm-level transparency	Implicit bias; latent structure; rank minimization	Geometric structure; class separability; curvature analysis; OT distribution analysis
Best Use Cases	Architecture comparison; understanding training dynamics	Building interpretable networks; signal processing tasks	Understanding implicit bias; low-rank structure	Analyzing representation geometry; class imbalance; generative model analysis; scientific discovery
Computational Cost	$O(n^2P)$ naive implementation; structured or approximate NTK computations can reduce this significantly (Novak et al. 2022)	Same as standard networks	Same as standard networks	Varies with manifold complexity
Key Papers	Jacot et al. (2018); Woodworth et al. (2020)	Gregor and Lecun (2010); Monga et al. (2021)	Razin and Cohen (2020); Razin et al. (2022)	Cohen et al. (2020); Ma et al. (2023); Lei et al. (2020); Labate and Shi (2024)

for large-scale models, yet these models most urgently need interpretability (Novak et al. 2022; Kiani et al. 2024).

6.1.2 Convergent insights across frameworks

Four recurring principles emerge across the classical frameworks reviewed here, suggesting common aspects of deep learning.

Implicit bias toward simplicity All frameworks reveal gradient descent's inherent preference for simple solutions. NTK shows convergence to minimum-norm interpolants (Jacot et al. 2018). Matrix factorization proves rank minimization bias (Razin and Cohen 2020). Sparse coding demonstrates preference for sparse representations (Papyan et al. 2017). Manifold learning reveals progressive dimensionality reduction (Cohen et al. 2020). Across all four frameworks gradient descent converges on related simplicity biases. This recurring pattern appears to reflect a general tendency of gradient-based optimization rather than coincidence.

Hierarchical learning as geometry Classical frameworks provide complementary geometric interpretations. Manifold analysis characterizes learning as untangling through radius,

dimensionality, and correlation reduction (Cohen et al. 2020). NTK spectral analysis reveals frequency-dependent learning rates (Cao et al. 2021). Sparse coding shows recruitment of increasingly abstract bases in deeper layers (Papyan et al. 2017). Matrix factorization demonstrates implicit locality bias (Razin et al. 2022). Together, these perspectives reveal representation learning as fundamentally geometric transformation from tangled high-dimensional distributions to linearly separable low-dimensional manifolds.

Architecture encodes inductive bias Classical theory rigorously connects architectural choices to task-relevant structure. NTK analysis proves CNNs with pooling exhibit different spectral properties than fully-connected networks (Xiao et al. 2020). Pooling operations create sparsity in information flow, selecting salient activations at each layer, analogous to how sparse PCA selects principal components, fundamentally changing the network's learning bias compared to dense fully-connected architectures. Sparse coding shows convolutional structures can be interpreted as hierarchical dictionary learning (Sulam et al. 2020). Matrix factorization establishes depth induces locality bias (Razin et al. 2022). Manifold learning proves depth enables smoother decision boundaries for manifold separation, acting as a fitting resource that controls the complexity of separating functions (Buchanan et al. 2021). These results formalize architecture design as principled encoding of domain knowledge through geometric and structural constraints.

Optimization shapes representations Training choices determine what networks learn, not just optimization speed. Large initialization produces networks that behave like kernel methods with fixed features (lazy regime), whereas small initialization enables rich feature learning (rich regime) (Woodworth et al. 2020). Large learning rates in gradient descent create different training dynamics and final solutions than gradual small-rate optimization (Ghosh et al. 2025). Strong sparsity penalties yield interpretable sparse codes, while weak penalties produce dense entangled features (Gregor and Lecun 2010). Training on imbalanced data causes networks to learn biased geometric representations favoring common classes (Ma et al. 2023). These optimization choices fundamentally shape which patterns emerge and how interpretable the learned representations are.

6.1.3 Mathematical connections between frameworks

Although the four classical frameworks have developed independently across disparate research communities, rigorous mathematical connections exist that reveal deeper unifying principles (Fig. 8).

Sparse representations and matrix factorization share the strongest mathematical connection through dictionary learning. Sparse coding decomposes data as $\mathbf{X} \approx \mathbf{DZ}$ where $\mathbf{Z}$ is sparse, while matrix factorization decomposes data as $\mathbf{X} \approx \mathbf{UV}$ with rank constraints. These formulations are mathematically equivalent where sparse coding is precisely matrix factorization with ℓ_1 regularization enforcing sparsity in the coefficient matrix. This explains why NMF appears in both frameworks (Zhang et al. 2021a; Xiong et al. 2022a) and why implicit regularization theory applies to both (Razin and Cohen 2020; Razin et al. 2021). The deep connection manifests practically in algorithm unrolling, where ISTA algorithms are transformed into neural networks through layer-wise matrix operations (Gregor and Lecun 2010; Monga et al. 2021).

NTKs and manifold learning both analyze learned geometry through complementary lenses. NTK characterizes learning dynamics through kernel geometry in function space, with spectral properties determining which functions networks learn efficiently (Cao et al. 2021; Xiao et al. 2020). Manifold learning characterizes the geometric transformations networks perform on data representations, revealing how classification capacity improves through coordinated reductions in manifold radius, dimensionality, and interclass correlations (Cohen et al. 2020). Both frameworks leverage eigendecomposition to analyze learned structure. NTK eigenvalues determine learning rates along different directions, whereas manifold analysis reveals intrinsic dimensionality of learned representations. Recent work has begun connecting these perspectives explicitly. Buchanan et al. (2021) showed that depth acts as a fitting resource enabling smoother manifold separation functions. This aligns with NTK analysis demonstrating that depth changes the kernel's spectral properties (Xiao et al. 2020).

Matrix factorization and manifold learning share fundamental dimensionality reduction principles. Matrix factorization's implicit bias toward low-rank solutions corresponds mathematically to manifold learning's principle that data lies on low-dimensional submanifolds ($d_{\mathcal{M}} \ll d_{\text{ambient}}$). The rank of the factorized representation directly relates to the intrinsic dimensionality of the data manifold. Both frameworks reveal that gradient descent implicitly discovers and exploits low-dimensional structure, with matrix factorization identifying this structure through rank minimization (Razin and Cohen 2020; Razin et al. 2021) and manifold learning through geometric compression (Cohen et al. 2020).

These mathematical connections suggest that the four frameworks are not entirely disparate methodologies but rather complementary lenses for examining the same fundamental phenomenon where neural networks learn low-dimensional interpretable structure through implicit regularization imposed by gradient descent dynamics. All four frameworks identify implicit bias toward simplicity. NTK biases toward minimum norm interpolants, sparse coding

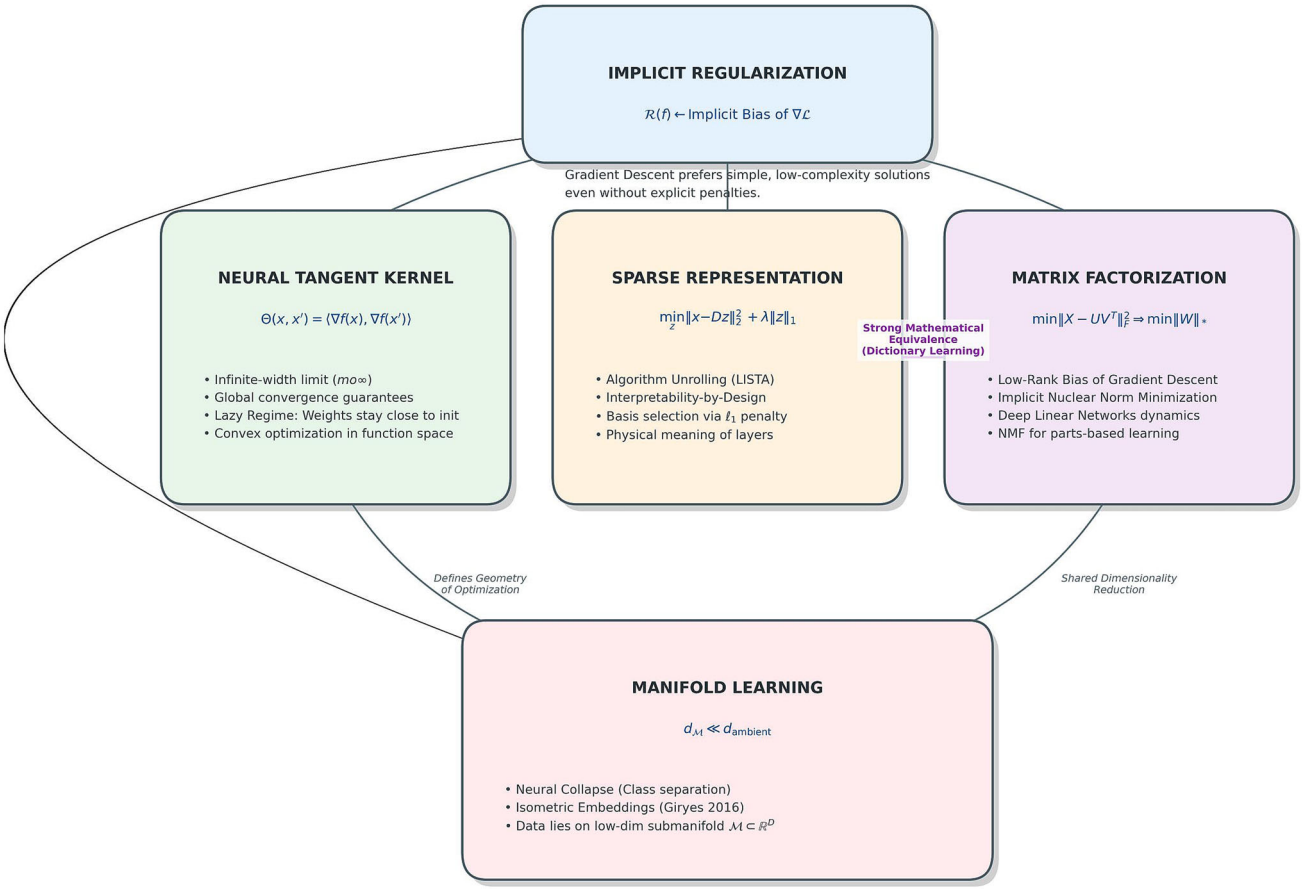


Fig. 8 Unified theoretical framework connecting classical ML perspectives on deep learning interpretability. All four frameworks share a common foundation in implicit regularization toward low-dimensional structure. We identify three rigorous mathematical bridges, (1) sparse representations and matrix factorization connect through dictionary learning (sparse coding is ℓ_1-constrained matrix factorization), (2) NTKs and manifold learning both analyze learned geometry through the kernel's spectral properties and manifold structure ($d_{\mathcal{M}} \ll d_{\text{ambient}}$), and (3) matrix factorization and manifold learning share dimensionality reduction principles where low-rank constraints correspond to intrinsic manifold dimensions. These connections reveal that classical frameworks represent complementary perspectives on the same underlying principle where neural networks learn interpretable structure through gradient-driven dimensionality reduction

toward sparse solutions, matrix factorization toward low-rank, and manifold learning toward low intrinsic dimension. This convergence across independently developed theoretical approaches suggests that gradient-based optimization often favors low-complexity structures that can support interpretability under appropriate validity conditions, offering partial theoretical foundations for why deep learning can be understood through classical frameworks despite operating in vastly over-parameterized regimes.

6.1.4 Framework-specific strengths

Each classical framework provides unique interpretability capabilities as discussed below.

- NTK provides training and generalization. NTK excels at characterizing when and why networks train successfully. Connecting neural networks to kernel regression enables convergence guarantees (Jacot et al. 2018), sample complexity bounds (Chen et al. 2020a), and learning priority predictions through spectral analysis (Cao et al. 2021). Applications demonstrate practical utility for physics-informed neural networks (McClenny and Braga-Neto 2023), meta-learning (Wang et al. 2022b), and federated learning (Huang et al. 2021). However, these insights require infinite width assumptions. Real networks with finite-width learn features through mechanisms beyond what kernel theory predicts, limiting practical applicability.
- Sparse representations provide interpretable architectures. Algorithm unrolling uniquely enables interpretable-by-design neural networks (Gregor and Lecun 2010; Monga et al. 2021). Mapping iterative sparse coding algorithms to network layers maintains explicit connections between operations and optimization objectives. Success across image restoration (Wang et al. 2020a), compres-

sive sensing (Li et al. 2024), HSI (Xu et al. 2022), and fault diagnosis (An et al. 2022) demonstrates that interpretable designs can match or approach black-box performance, challenging assumed interpretability-accuracy trade-offs.

- Matrix factorization provides implicit regularization. Matrix factorization theory illuminates fundamental optimization properties. The finding that gradient descent in matrix factorization can implicitly promote low-rank solutions, in ways not fully captured by standard norm minimization (Razin and Cohen 2020), has reshaped understanding of generalization in over-parameterized models. Extensions to tensor (Razin et al. 2021) and hierarchical factorization (Razin et al. 2022) reveal how depth induces locality bias in CNNs. Connections to copositive programming enable polynomial-time global optimization in certain two-layer networks (Sahiner et al. 2021). NMF provides concept-based explanations through parts-based decomposition (Zhang et al. 2021a; Parekh et al. 2024).
- Manifold learning provides geometric interpretability. Manifold learning offers geometric tools unavailable through other frameworks. Characterizing classification capacity through manifold radius, dimensionality, and correlations (Cohen et al. 2020) provides mathematical foundations for understanding representation learning. Curvature analysis reveals that curvature imbalance causes long tail bias (Ma et al. 2023). Geometric transformer analysis correlates with forecasting performance (Kaufman and Azencot 2024). Adversarial regions explain adversarial examples as off-manifold perturbations (Zhang et al. 2020a).

6.1.5 The fundamental tension

Across all four frameworks, the gap between idealized assumptions and real-world models is the recurring source of limitation. The simplifying conditions that make each theory tractable (infinite width for NTK, exact sparsity for sparse coding, linear or tensor structure for matrix factorization, and the manifold hypothesis for manifold learning) are precisely the conditions that practical networks violate. The per-section Critical Analysis subsections and the *Validity Conditions*, *Gap to Practice*, and *Failure Modes* rows of Table 8 document these gaps framework by framework, and here we draw them together into a single cross-cutting observation.

A fundamental tension pervades classical approaches to interpretability. Theoretical rigor inversely correlates with practical applicability. Methods with strongest mathematical guarantees (NTK theory in infinite-width limit) apply to idealized settings diverging from practice. Methods with broadest practical applicability (algorithm unrolling, sparse autoencoders) sacrifice some theoretical rigor for utility. This tension is not accidental but inherent because mathematical tractability requires simplifying assumptions (infinite width, linear networks, idealized training) that eliminate precisely the complexities (finite width, nonlinearity, stochastic optimization) making deep learning effective.

This creates situations where theoretical predictions diverge from practical behavior. Theory suggests infinite-width networks cannot benefit from depth beyond kernel composition, yet practical deep networks dramatically outperform shallow ones through hierarchical feature learning. Theory characterizes lazy regime where features barely change, yet empirical success stems from rich feature learning. Theory assumes exact gradient descent, yet practice uses noisy stochastic gradients with momentum and adaptive rates.

Resolving this tension requires either (1) accepting that theory provides approximate understanding rather than precise predictions, or (2) developing fundamentally new theoretical frameworks that maintain rigor while accommodating practical complexities. The field has pursued both approaches. Some researchers extend NTK to finite width (Seleznova and Kutyniok 2022a) and feature learning (Allen-Zhu et al. 2019; Chen et al. 2020a), developing approximate theories that accept reduced precision for broader applicability. Others explore optimization dynamics beyond classical analysis (Ghosh et al. 2025; Allen-Zhu and Li 2023), seeking entirely new theoretical frameworks.

6.2 Final remarks

6.2.1 The necessity of theory

Deep learning has achieved remarkable empirical success, mastering games (Silver et al. 2016), protein folding (Jumper et al. 2021), and conversational AI (ChatGPT, Gemini), yet this success rests on foundations we incompletely understand. Trial and error tuning, though effective, cannot indefinitely sustain progress. As models grow in scale, cost, and societal impact, principled understanding becomes imperative. In high stakes domains such as medical diagnosis, autonomous vehicles, and criminal justice, deployment without interpretability raises fundamental questions of trust and accountability. Medical professionals may be reluctant to rely on deep learning based clinical decision support systems for cancer diagnosis when these systems cannot explain their reasoning, even if empirical accuracy appears high.

Classical ML theory provides that principled foundation. Kernel methods, sparse representations, matrix factorization, and manifold learning offer rigorous mathematical frameworks for analyzing deep learning through lenses of functional analysis, convex optimization, linear algebra, and differential geometry. These frameworks connect deep learning to centuries of mathematical knowledge, enabling

borrowing of analytical techniques and transfer of theoretical insights.

However, current theory is incomplete. The lazy-rich divide, idealized infinite-width assumptions, and computational intractability prevent complete theoretical understanding of practical deep learning. This incompleteness does not negate theory's value. Partial understanding rigorously established surpasses complete ignorance or unfounded speculation. Classical frameworks provide among the most rigorous available mathematical foundations for interpretability, even while acknowledging significant gaps.

6.2.2 Beyond black box

The field faces a choice between accepting deep learning as empirical craft based on heuristics and intuitions, or developing it as rigorous science grounded in mathematical theory. Classical frameworks represent commitment to the latter path, pursuing mathematical understanding even when simplifying assumptions are required, developing rigorous methods even when computational costs are high, seeking provable guarantees even when approximations are necessary.

This commitment yields concrete benefits. Algorithm unrolling demonstrates that theory-guided architecture design can produce interpretable yet high-performing systems in several domains (Monga et al. 2021). NTK spectrum analysis can help predict aspects of trainability and generalization through efficient kernel computation, reducing reliance on expensive full training of architectures likely to fail (Xiao et al. 2020). Implicit regularization theory explains why deep learning generalizes, informing regularization and optimization choices (Razin and Cohen 2020; Razin et al. 2022). Manifold learning provides geometric interpretability, characterizing how networks can transform data structure through coordinated changes in radius, dimensionality, and correlations under the manifold hypothesis (Cohen et al. 2020).

Theory's value extends beyond immediate applications. Theoretical frameworks organize knowledge systematically, suggest new experiments, and constrain what is possible. This enables cumulative understanding that transcends individual observations. Even when specific predictions prove inaccurate, the conceptual frameworks and analytical techniques remain valuable, enabling refinements that progressively improve our understanding. Scientific fields with strong theoretical foundations progress more systematically than those relying primarily on empirical exploration.

6.2.3 The path forward

Classical ML theory has established firm foundations for deep learning interpretability while revealing profound limitations requiring future work. The path forward requires balanced investment across multiple priorities. Theoretical advances addressing finite width dynamics, unified implicit regularization, and sparse representation theory for transformers will strengthen mathematical foundations. Methodological development of efficient computation, standardized benchmarks, and hybrid approaches will enhance practical applicability. Bridge building connecting classical theory to mechanistic interpretability, causal inference, and neuroscience will yield richer understanding through cross pollination. Emerging directions in foundation model interpretability, multimodal learning, generative models, and trustworthy AI will extend classical frameworks to modern systems. Progress across these fronts requires sustained commitment to high risk theoretical work, development of practical tools, and interdisciplinary collaboration advancing interpretability research.

6.2.4 Toward interpretable by design systems

We envision a future where deep learning systems are fundamentally interpretable, not through post hoc rationalizations grafted onto opaque models, but through architectural designs that build interpretability in from inception. Algorithm unrolling (Monga et al. 2021) provides an existence proof that such interpretable-by-design systems can approach black-box performance, with demonstrations across image restoration (Wang et al. 2020a) and compressive sensing (Li et al. 2024) showing competitive or superior results. Extending this paradigm across domains and architectures could establish a new interpretability standard where explanations come with mathematical guarantees rather than heuristic plausibility arguments.

Realizing this vision requires advancing both theoretical foundations and practical implementations. Classical ML theory provides essential mathematical frameworks, but must be extended to handle modern architectures, training regimes, and application domains. The challenge lies in developing interpretable by design approaches that scale to contemporary systems while maintaining rigorous guarantees. Progress will depend on continued development of theory guided architectures, efficient computational methods, and empirical validation demonstrating that interpretability and performance can coexist.

6.2.5 Closing reflection

We have systematically examined how classical ML frameworks establish theoretical foundations for transforming opaque black boxes into reliable components of knowledge-based systems. Although current theory remains incomplete, constrained by infinite width assumptions and architectural restrictions, these limitations delineate the research frontier rather than fundamental barriers. By identifying validity con-

ditions and fundamental limitations across kernel methods, sparse representations, matrix factorization, and manifold learning, this review provides practitioners with a roadmap for selecting appropriate theoretical frameworks when building AI-driven decision support systems. Every gap between theory and practice represents an opportunity for theoretical advancement that can enhance the reliability of knowledge extraction from deep neural networks.

The journey from empirical craft to rigorous science is ongoing. Classical frameworks have established essential foundations, much as classical physics provided the mathematical scaffolding for quantum mechanics despite its ultimate conceptual inadequacy at atomic scales. By building on these foundations, addressing identified limitations, and developing new theoretical tools, the field can progressively illuminate deep learning's remaining mysteries. This review synthesizes the mathematical foundations that enable practitioners to build interpretable-by-design systems supporting transparent, trustworthy decision-making in high-stakes knowledge-based applications. The analytical frameworks exist, the challenges are clear, and the path forward awaits systematic exploration toward knowledge-based AI systems where decisions are both accurate and explainable.

Author Contributions M.U.K. conceived the study, designed the methodology, implemented the work, and wrote the main manuscript. A.A.K.A. handled project administration and supervision. S.B. performed methodology validation, prepared the figures, and reviewed and edited the manuscript. S.R. reviewed and edited the manuscript. All authors reviewed the manuscript.

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Declarations

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